MODEL THEORY AND THE BELL–KOCHEN–SPECKER THEOREM

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Abstract. We give a countable theory of real-valued $L_{\omega_1,\omega}$ whose standard models are the lattices of closed subspaces of separable Hilbert spaces with quantum measures, so that Gleason's theorem holds in them, and no model of which carries a two-valued measure. Stating the model existence theorem as a game, we show that this impossibility is witnessed by a well-founded tree of finite obstructions, which branches infinitely only at countable suprema and infima.

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Introduction

Gleason's theorem and the Bell–Kochen–Specker theorem are statements about all quantum measures on all Hilbert spaces. We show that both follow from a countable theory T of the real-valued $L_{\omega_1,\omega}$ of the first author [Aub], and that the second is witnessed by finite obstructions, although the logic is not compact. The theory has a Hilbert sort, a Solèr sort for the lattice, a sort of square-summable sequences of vectors, and a real-valued predicate

μ on the Solèr sort; a model is *standard* when its two vector sorts are the closed unit balls of Hilbert spaces (see Definition 5.2). For an element a of the Solèr sort, S_a is the closed subspace spanned by the vectors that belong to a, and P_a is the orthogonal projection onto S_a (see Section 5.2).

Theorem A (Gleason's theorem in real-valued $L_{\omega_1,\omega}$). *There is a countable theory T of real-valued $L_{\omega_1,\omega}$ with the following two properties.*

(1) In every standard model of T, the Hilbert sort is the closed unit ball of a separable infinite-dimensional complex Hilbert space H, the map $a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort onto the lattice $L(H)$ of closed subspaces of H, and there is a unique density operator ρ on H such that $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort.

(2) Conversely, for every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, there is a standard model of T whose Hilbert sort is the closed unit ball of H, whose Solèr sort is $L(H)$, and in which $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in L(H)$.

Theorem B (The range of the measure). *Let T be the theory of Theorem A. In every standard model of T, the range of μ is the whole interval $[0,1]$. Consequently, for every Borel set $B \subseteq \mathbb{R}$ that meets $[0,1]$, the theory T together with the axiom*

$$\sup_a \mathbf{1}_B(\mu(a)) = 0,$$

which says that no value of μ lies in B, has no standard models; here $\mathbf{1}_B(\mu(a))$ is the formula of Lemma 5.1, whose value at a is 1 when $\mu(a) \in B$ and 0 otherwise.

Theorem C (The Bell–Kochen–Specker theorem in real-valued $L_{\omega_1,\omega}$). *The theory T of Theorem A, together with the axiom $\min(\mu(a), 1-\mu(a)) = 0$, which says that μ takes only the values 0 and 1, has no models, standard or not.*

Theorem D (Finite obstructions). *Let T' be a countable theory of real-valued $L_{\omega_1,\omega}$ that has no models. In the model existence game of Section 10, the player who claims that T' has no models has a winning strategy. The tree of plays of every such strategy is well-founded, its leaves are finite sets of commitments satisfied in no structure, and it branches infinitely only at countable suprema and infima; once the answers there are fixed, the plays have bounded length. For T' the theory of Theorem C, every finite subset of T' has a model, and the tree of every winning strategy branches infinitely.*

For the theory of Theorem C, the proof of that theorem leads to a winning strategy whose leaves contain configurations of the Kochen–Specker type, one block of four basis vectors at a time (see Figure 12).

Roadmap. Sections 1–4 are background, and Section 4 is optional. In Sections 5 and 6 we present the logic and the theory T, and in Sections 7–9 we prove Theorems A, B and C. In Section 10 we introduce the game and prove Theorem D; the remarks of Section 11 are optional.

1. Lattices

Birkhoff and von Neumann [BvN36] observed that the experimentally verifiable propositions about a quantum system correspond to the closed subspaces of a Hilbert space, and that these form a lattice in which the distributive law fails; the logic of quantum mechanics is therefore not that of a Boolean algebra. They proposed the weaker modular law in its place, which holds in finite dimensions but fails for the closed subspaces of an infinite-dimensional Hilbert space. We begin with the lattice-theoretic terminology that this observation calls for.

Definition 1.1. A *partially ordered set* is a set with a reflexive, antisymmetric and transitive relation $\leq$. For a subset S of a partially ordered set, an *upper bound* of S is an element b with $a \leq b$ for all $a \in S$, and the *join* $\bigvee S$ of S is its least upper bound, when it exists; lower bounds and the *meet* $\bigwedge S$ are defined dually. A *lattice* is a partially ordered set in which every pair of elements a, b has a join $a \vee b$ and a meet $a \wedge b$. A lattice is *bounded* when it has a least element 0 and a greatest element 1, and *complete* when every subset has a join.

In a lattice, the relation $a \leq b$ holds if and only if $a \wedge b = a$; moreover, a lattice is complete if and only if every subset has a meet, since the meet of a subset is the join of its lower bounds.

Definition 1.2. An *orthocomplementation* on a bounded lattice Λ is a map $a \mapsto a^\perp$ from Λ to Λ such that, for all $a, b \in \Lambda$, the inequality $a \leq b$ implies $b^\perp \leq a^\perp$, and $a^{\perp\perp} = a$, $a \wedge a^\perp = 0$ and $a \vee a^\perp = 1$. A bounded lattice with an orthocomplementation is an *ortholattice*. Two elements a, b of an ortholattice are *orthogonal*, written $a \perp b$, when $a \leq b^\perp$. An ortholattice is *orthomodular* when, for all a, b, the inequality $a \leq b$ implies $b = a \vee (b \wedge a^\perp)$.

In $L(H)$, orthomodularity says that every closed subspace N containing a closed subspace M is the orthogonal sum of M and $N \cap M^\perp$. In an ortholattice, the De Morgan law $(\bigvee S)^\perp = \bigwedge\{a^\perp : a \in S\}$ holds whenever the join exists.

Definition 1.3. An *atom* of a bounded lattice is an element $p \neq 0$ such that $b \leq p$ implies $b = 0$ or $b = p$. A bounded lattice is *atomistic* when every element a is the join of the atoms below it (this join being required to exist). It has the *covering property* when, for every atom p and every element a with $p \not\leq a$, no element lies strictly between a and $a \vee p$. It is *irreducible* when, for any two distinct atoms p and q, some atom r distinct from p and q lies below $p \vee q$.

For complete atomistic lattices with the covering property, irreducibility in this sense is equivalent to the absence of a decomposition as a product of two lattices with more than one element (see Maeda and Maeda [MM70]); the definition above is the one that the coordinatization theorem of Section 3 requires.

Definition 1.4. A *sub-ortholattice* of an ortholattice Λ is a subset that contains 0 and 1 and is closed under the operations $\vee$, $\wedge$ and $^\perp$ of Λ. An *embedding* of ortholattices is an injective map that preserves $\vee$, $\wedge$, $^\perp$, 0 and 1.

We shall also need a notion of density that concerns the order alone; no topology is involved. Let Γ be a complete lattice and Λ a subset of Γ. We say that Λ is *join-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigvee\{a \in \Lambda : a \leq g\}$, the join being computed in Γ, and that Λ is *meet-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigwedge\{a \in \Lambda : a \geq g\}$. For instance, $\mathbb{Q}$ is join-dense and meet-dense in the extended real line $[-\infty, \infty]$. For a partially ordered set Λ and $X \subseteq \Lambda$, let X^u and X^l be the sets of upper and lower bounds of X in Λ; a *cut* is a subset

X with $X = (X^u)^l$. The cuts, ordered by inclusion, form the *MacNeille completion* $\mathrm{DM}(\Lambda)$ of Λ [Mac37] (also called the Dedekind–MacNeille completion, which explains the notation), a complete lattice into which $a \mapsto \{b \in \Lambda : b \leq a\}$ embeds Λ, preserving all joins and meets that exist in Λ.

Proposition 1.5 (Banaschewski [Ban56], Schmidt [Sch56]). *Let Λ be a partially ordered set, and let Γ be a complete lattice that contains Λ as a join-dense and meet-dense subset, its order extending that of Λ. For $g \in \Gamma$, put $\Lambda_{\leq g} = \{a \in \Lambda : a \leq g\}$. Then $g \mapsto \Lambda_{\leq g}$ is an isomorphism of Γ onto $\mathrm{DM}(\Lambda)$ that restricts to the canonical embedding on Λ. If Λ is an ortholattice, an orthocomplementation of Γ that extends that of Λ is unique, and it is given by $g^\perp = \bigwedge\{a^\perp : a \in \Lambda,\ a \leq g\}$.*

Proof. By join-density, we have $g = \bigvee \Lambda_{\leq g}$; hence the upper bounds of $\Lambda_{\leq g}$ in Λ are the elements $b \in \Lambda$ with $b \geq g$. By meet-density, their common lower bounds are the elements $a \in \Lambda$ with $a \leq g$, and $\Lambda_{\leq g}$ is a cut. Join-density also shows that $g \leq g'$ if and only if $\Lambda_{\leq g} \subseteq \Lambda_{\leq g'}$. If X is a cut and $g = \bigvee X$ in Γ, then X is contained in $\Lambda_{\leq g}$; conversely, an element $a \in \Lambda$ with $a \leq g$ lies below every $b \in X^u$, since each such b is an upper bound of X in Γ, and therefore $a \in X = (X^u)^l$. The formula for $g^\perp$ follows from $g = \bigvee \Lambda_{\leq g}$ and the De Morgan law. ☺

2. Hermitian spaces

The coordinatization theorem, which we state in Section 3, turns the atoms of a lattice into the one-dimensional subspaces of a vector space over a division ring, and orthocomplementation into orthogonality with respect to a form. In this section we collect the facts about such spaces that we need; the decisive one is that orthomodularity of the lattice of orthogonally closed subspaces is a property of the space itself (see Proposition 2.8).

Definition 2.1. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is invertible; multiplication need not be commutative. An *involution* of a division ring Δ is a map $\alpha \mapsto \alpha^*$ such that $(\alpha + \beta)^* = \alpha^* + \beta^*$, $(\alpha\beta)^* = \beta^*\alpha^*$ and $\alpha^{**} = \alpha$ for all $\alpha, \beta \in \Delta$. Let E be a left vector space over Δ. A *hermitian form* on E is a map $\langle \cdot \mid \cdot \rangle \colon E \times E \to \Delta$ such that $\langle \alpha x + \beta y \mid z \rangle = \alpha\langle x \mid z \rangle + \beta\langle y \mid z \rangle$ and $\langle y \mid x \rangle = \langle x \mid y \rangle^*$ for all $x, y, z \in E$ and $\alpha, \beta \in \Delta$; it is *anisotropic* when $\langle x \mid x \rangle = 0$ implies $x = 0$. A *hermitian space* is a left vector space over a division ring with an involution, together with an anisotropic hermitian form.

It follows that $\langle x \mid \alpha y \rangle = \langle x \mid y \rangle \alpha^*$. The bracket is linear in its first argument, as is customary in mathematics; in Dirac notation, which physicists use, it is linear in the second. Over $\mathbb{Q}$ with the identity involution, the form $\sum_n x_n y_n$ on finitely supported sequences is anisotropic, since a sum of rational squares vanishes only when every square does; nothing in the definition forces the scalars to be real or complex.

Definition 2.2. In a hermitian space E, two vectors x and y are *orthogonal*, written $x \perp y$, when $\langle x \mid y \rangle = 0$. For $S \subseteq E$, let $S^\perp$ be the set $\{x \in E : \langle x \mid y \rangle = 0 \text{ for all } y \in S\}$. A subspace M is *orthogonally closed* when $M^{\perp\perp} = M$.

Orthogonality is symmetric, since $\langle y \mid x \rangle = \langle x \mid y \rangle^*$.

Definition 2.3 (The lattice $\mathcal{C}(E)$). For a hermitian space E, we write $\mathcal{C}(E)$ for the set of orthogonally closed subspaces of E, ordered by inclusion.

Every set of the form $S^\perp$ is a subspace, and it is orthogonally closed, since $S^{\perp\perp\perp} = S^\perp$; intersections of orthogonally closed subspaces are orthogonally closed. Hence $\mathcal{C}(E)$ is a complete lattice with least element $\{0\}$ and greatest element E; the meet of a family $(M_j)_{j \in J}$ is $\bigcap_j M_j$, and its join is $\big(\sum_j M_j\big)^{\perp\perp}$, which may be strictly larger than $\sum_j M_j$. For

$M \in \mathcal{C}(E)$, anisotropy gives $M \cap M^\perp = \{0\}$, whence $M \vee M^\perp = (M^\perp \cap M)^\perp = E$; thus $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is a complete ortholattice.

Definition 2.4. A hermitian space E is *orthomodular* when $M + M^\perp = E$ for every $M \in \mathcal{C}(E)$, where $M + M^\perp$ is the set of sums $v + y$ with $v \in M$ and $y \in M^\perp$.

The lattice join $M \vee M^\perp$ is always E; orthomodularity asks that the algebraic sum be E as well.

Definition 2.5. A *Hilbert space* is a hermitian space H over the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ or the quaternions $\mathbb{H}$, with the usual involution (the identity, complex conjugation or quaternionic conjugation), such that $\langle x \mid x\rangle > 0$ for $x \neq 0$ and H is complete in the norm $\|x\| = \langle x \mid x\rangle^{1/2}$.

For a Hilbert space H, the orthogonally closed subspaces are exactly the norm-closed subspaces; thus $\mathcal{C}(H)$ is the lattice $L(H)$ of closed subspaces of H, and it is orthomodular by the projection theorem, which gives $M + M^\perp = H$ for every closed subspace M.

Lemma 2.6. *In a hermitian space E, every finite-dimensional subspace F is orthogonally closed and satisfies $F + F^\perp = E$.*

Proof. By anisotropy, the Gram–Schmidt process gives F an orthogonal basis $w_1, \dots, w_n$ with $\langle w_i \mid w_i\rangle \neq 0$ for $i = 1, \dots, n$. For $z \in E$, the vector $Pz = \sum_i \langle z \mid w_i\rangle \langle w_i \mid w_i\rangle^{-1} w_i$ lies in F, and $z - Pz$ lies in $F^\perp$; hence $F + F^\perp = E$. If $y \in F^{\perp\perp}$, the vector $y - Py$ lies in $F^\perp \cap F^{\perp\perp}$ and is therefore orthogonal to itself; by anisotropy it is 0, whence $y \in F$. ☺

Lemma 2.7. *Let E be a hermitian space over Δ. If $M \in \mathcal{C}(E)$ and $x \in E$, then $M + \Delta x \in \mathcal{C}(E)$.*

Proof. If $x \in M$, there is nothing to prove; we therefore assume that $x \notin M$. Put $W = M^\perp$, and let ξ_x be the functional $z \mapsto \langle z \mid x\rangle$ on W; it is nonzero, since $x \notin W^\perp = M$, and we may choose $z_0 \in W$ with $\xi_x(z_0) = 1$. Let y be an element of $(M + \Delta x)^{\perp\perp}$, and let ξ_y be the functional $z \mapsto \langle z \mid y\rangle$ on W. Since $W \cap x^\perp = (M + \Delta x)^\perp$, the functional ξ_y vanishes on the kernel of ξ_x; as $z - \xi_x(z) z_0$ lies in that kernel for every $z \in W$, we obtain $\xi_y(z) = \xi_x(z)\alpha$ with $\alpha = \xi_y(z_0)$. Because $\langle z \mid \alpha^* x\rangle = \langle z \mid x\rangle \alpha$, the vector $y - \alpha^* x$ is orthogonal to W and therefore lies in $W^\perp = M$. Thus $y \in M + \Delta x$. ☺

Proposition 2.8. *Let E be a hermitian space. The lattice $\mathcal{C}(E)$ is orthomodular if and only if E is an orthomodular space.*

Proof. Suppose first that E is orthomodular, and let $M \subseteq M'$ be elements of $\mathcal{C}(E)$. Every vector $z \in M'$ can be written as $v + y$ with $v \in M$ and $y \in M^\perp$; since $y = z - v$ lies in M', we have $M' = M + (M' \cap M^\perp) \subseteq M \vee (M' \wedge M^\perp) \subseteq M'$, which is the orthomodular law. Conversely, suppose that $\mathcal{C}(E)$ is orthomodular, let M be an element of $\mathcal{C}(E)$, and let x be a vector of $E \setminus M$. By Lemma 2.7, the subspace $M' = M + \Delta x$ belongs to $\mathcal{C}(E)$, and orthomodularity gives $M' = M \vee (M' \cap M^\perp)$. Since $M' \neq M$, the subspace $M' \cap M^\perp$ is nonzero; as it meets M only in 0, while M has codimension one in M', it is a one-dimensional subspace Δy. Then $M \subsetneq M + \Delta y \subseteq M'$, and the codimension count gives $M' = M + \Delta y$; hence $x \in M + M^\perp$. ☺

3. Projective geometry, Solèr lattices and pre-Solèr lattices

In this section we assume familiarity with Hilbert spaces, but none with projective geometry. Our purpose is to explain how an ortholattice with a few combinatorial properties can be represented by orthogonally closed subspaces of a vector space with a hermitian form, and how Solèr's condition forces the scalars to be real, complex or quaternionic.

3.1. **Points and lines.** In a complex Hilbert space H, the atoms of $L(H)$ are the one-dimensional subspaces $\mathbb{C}x$, with $x \neq 0$, called *rays*; the join of two distinct rays is the two-dimensional subspace, or *plane*, that they span. More generally, we call the one-dimensional subspaces of any vector space its *rays*. Which rays lie in which planes is information about the lattice alone (see Figure 1); the theorems below show that it determines the space up to the choice of scalars.

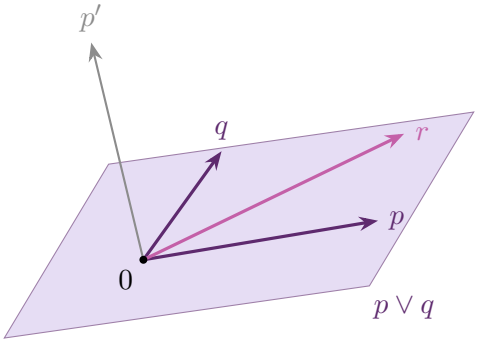


FIGURE 1. Atoms of $L(H)$ are rays. The join $p \vee q$ of two rays is the plane they span; the ray r in that plane is an atom below $p \vee q$, while the ray p' is not.

Definition 3.1. A *projective geometry* is a set of *points* together with a family of subsets, called *lines*, such that:

(PG1) any two distinct points lie on exactly one line;
(PG2) every line contains at least three points;
(PG3) (*the Veblen–Young axiom*) if a, b, a', b' are distinct points and the line through a and b meets the line through a' and b', then the line through a and a' meets the line through b and b'.

A set of points is a *projective subspace* when it contains the whole line through any two of its points; the projective subspace *generated* by a set of points is the intersection of all projective subspaces containing it, and a *projective plane* is the projective subspace generated by three points not on one line. Axiom (PG3) says that two lines lying in a common projective plane always meet (see Figure 2); a projective geometry has no parallel lines. The name comes from perspective drawing, in which parallel rails, projected onto the canvas, meet at the horizon.

The basic example comes from linear algebra. Let V be a left vector space over a division ring Δ, of dimension at least two. Take as points the rays of V, and as lines the sets of rays contained in a two-dimensional subspace. Axiom (PG1) holds because two distinct rays span a unique two-dimensional subspace, and axiom (PG2) because the two-dimensional subspace spanned by linearly independent vectors x and y contains the three distinct rays Δx, Δy and $\Delta(x + y)$. For (PG3), if the two-dimensional subspaces spanned by a, b and by a', b' have a ray in common, then a, b, a', b' lie in a subspace of dimension at most three, in which any two two-dimensional subspaces have a ray in common, by counting dimensions.

Points are *independent* when none of them lies in the projective subspace generated by the others; in the basic example, rays are independent exactly when nonzero vectors spanning them are linearly independent. The following theorem says that, as soon as there are four independent points, the basic example is the only one.

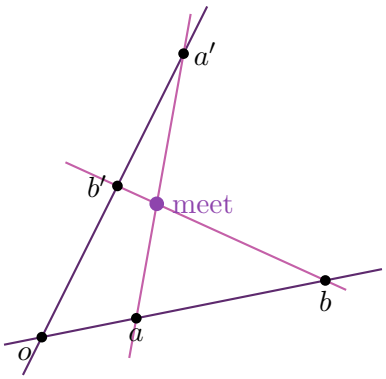


FIGURE 2. The Veblen–Young axiom. The lines ab and $a'b'$ meet at o; then the lines aa' and bb' meet as well.

Theorem 3.2 (Coordinatization of projective geometries; see Baer [Bae52]). *Every projective geometry that contains four independent points is isomorphic, by a bijection of points that maps lines onto lines, to the geometry of rays of a left vector space E over a division ring Δ. The pair (Δ, E) is unique up to a* semilinear isomorphism*: an additive bijection between the two vector spaces that is linear except that it moves scalars by an isomorphism between the two division rings.*

Two features of this theorem matter below: the division ring Δ is recovered from the geometry alone, and it can be any division ring, whether the rational numbers, a non-archimedean ordered field (one with infinitesimals, such as a field of formal Laurent series) or the quaternions; and the four independent points cannot be dispensed with. A projective plane, regarded as a projective geometry in its own right, need not come from a division ring: those that do are exactly the planes in which Desargues's theorem holds (if two triangles are in perspective from a point, the three intersection points of corresponding sides lie on one line), and the others, the *non-Desarguesian* planes, exist. In a geometry with four independent points, Desargues's theorem follows from the axioms.

3.2. **From lattices to geometries.** Let Λ be an atomistic lattice with the covering property. We take its atoms as the points of a geometry whose lines are the sets $\{r : r \text{ an atom with } r \leq p \vee q\}$, for distinct atoms p and q; in $L(H)$, these are the rays in the plane spanned by p and q. The covering property makes this geometry satisfy (PG1) and (PG3), and irreducibility in the sense of Definition 1.3 is exactly (PG2); see Maeda and Maeda [MM70, §§33–34] and Holland [Hol95, §3].

Orthocomplementation adds a relation between points: by the definition in Section 1, atoms p and q are orthogonal when $p \leq q^{\perp}$. In $L(H)$ this is orthogonality of rays, and it determines the inner product up to the ambiguity described at the end of Theorem 3.3 below. Consider the following properties of an ortholattice Λ:

(C) the lattice is complete;
(OM) it is orthomodular;
(A) it is atomistic;
(CP) it has the covering property;
(I) it is irreducible;
(D) it contains four mutually orthogonal atoms.

The following theorem, which goes back to Birkhoff and von Neumann [BvN36] in finite dimensions, says that orthogonality of atoms determines a form in the same way for every ortholattice with (OM), (A), (CP), (I) and (D). Condition (D) supplies the four independent points of Theorem 3.2, since mutually orthogonal atoms are independent.

Theorem 3.3 (Coordinatization; Baer [Bae52], Maeda and Maeda [MM70, §34], Holland [Hol95, §3]). *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I) and (D), but not necessarily (C). Then there are a division ring Δ with an involution, a hermitian space E over Δ, and a bijection $p \mapsto R_p$ from the atoms of Λ onto the rays of E, such that two atoms p and q are orthogonal exactly when $R_p \perp R_q$. The map*

$$\gamma\colon \Lambda \to \mathcal{C}(E), \qquad \gamma(a) = \bigcup\{R_p : p \text{ an atom with } p \le a\},$$

is an embedding of ortholattices whose image contains every finite-dimensional subspace of E and the orthogonal complement of each. The triple formed by Δ, E and the form is unique up to a semilinear isomorphism followed by a rescaling $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta$, where $\zeta^* = \zeta \neq 0$, with the new involution $\alpha \mapsto \zeta^{-1}\alpha^*\zeta$.

Proof of the statements about γ. The orthocomplementation of Λ induces a *polarity* on the geometry of its atoms: an inclusion-reversing bijection of the projective subspaces onto themselves, whose square is the identity, which sends each projective subspace to the set of points orthogonal to all its points. By Theorem 3.2, the atoms become the rays of a vector space E, and by Baer's theorem [Bae52], every polarity of this kind under which no point is orthogonal to itself is orthogonality for an anisotropic hermitian form on E. If two atoms lie below a, every atom on the line they determine lies below a; hence $\gamma(a)$ is a subspace. For an atom r, the following are equivalent: $r \le a^\perp$; $a \le r^\perp$; $p \le r^\perp$ for every atom $p \le a$ (because a is the join of these atoms, by (A)); $R_r \perp R_p$ for every atom $p \le a$. Thus $\gamma(a^\perp) = \gamma(a)^\perp$, and applying this twice gives $\gamma(a) = \gamma(a)^{\perp\perp}$. By (A), the relation $a \le b$ holds if and only if $\gamma(a) \subseteq \gamma(b)$; hence γ is injective. Meets are preserved because an atom lies below $a \wedge b$ exactly when it lies below a and b, and joins are preserved by the De Morgan law. If $x_1, \dots, x_k$ is a basis of a finite-dimensional subspace and p_i is the atom with $R_{p_i} = \Delta x_i$, the subspace is $\gamma(p_1 \vee \cdots \vee p_k)$, the image of a finite join, which exists in every lattice; its orthogonal complement is $\gamma\big((p_1 \vee \cdots \vee p_k)^\perp\big)$. ☻

Piron [Pir64] proved this representation for complete lattices; completeness is not needed for the embedding, and in Theorem 3.6 below it serves only to make γ surjective.

3.3. **Harmonic conjugates and Solèr's condition.** The coordinatization leaves the division ring Δ undetermined. Solèr's condition singles out the real, complex and quaternionic numbers, and it is phrased through a classical construction of projective geometry.

Let p and q be distinct points, write $p + q$ for the line through them (in the basic example, the rays of the plane spanned by p and q), and let c be a point of $p + q$ other than p and q. Two distinct lines in a common projective plane meet in exactly one point, by (PG3), and we denote that point by an intersection sign. Choose a point r not on $p + q$ and a point r' on $r + p$ other than r and p; put $c' = (c + r') \cap (q + r)$ and $c'' = (p + c') \cap (q + r')$. The point $h = (r + c'') \cap (p + q)$ does not depend on the choices of r and r', as the computation below shows; it is the *harmonic conjugate* of c with respect to p and q (see Figure 3). In a lattice satisfying (A), (CP) and (I), the same construction applies to atoms, with $\vee$ in place of $+$ and $\wedge$ in place of $\cap$; the relation "h is the harmonic conjugate of c with respect to p and q" is then expressed by a first-order formula in the language of lattices.

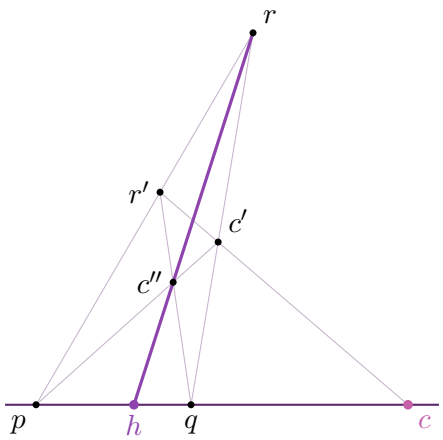


FIGURE 3. The harmonic conjugate h of c with respect to p and q.

In coordinates the construction is simple. Let u and v be vectors spanning p and q, scaled so that $c = \Delta(u + v)$, and let w be a vector spanning r, scaled so that $r' = \Delta(u + w)$, which is possible because r' lies on $r + p$ and differs from r and p. A direct computation gives $c' = \Delta(v - w)$, $c'' = \Delta(u - v + w)$ and $h = \Delta(u - v)$. All coefficients are ± 1, so the computation is valid over every division ring:

$$\text{the harmonic conjugate of } \Delta(u+v) \text{ with respect to } \Delta u \text{ and } \Delta v \text{ is } \Delta(u-v). \tag{1}$$

Now suppose that $p \perp q$; then $u \perp v$. The orthocomplement of c within the plane $p \vee q$, the ray $c^\perp \wedge (p \vee q)$, consists of the vectors $\alpha u + \beta v$ with $\langle \alpha u + \beta v \mid u + v \rangle = \alpha \langle u \mid u \rangle + \beta \langle v \mid v \rangle = 0$. It coincides with the harmonic conjugate $\Delta(u - v)$ exactly when $\langle u \mid u \rangle = \langle v \mid v \rangle$, that is, when u and v have the same "length" (see Figure 4). In characteristic 2, that is, when $1 + 1 = 0$ in Δ, the vectors $u + v$ and $u - v$ coincide, and the equality $\langle u \mid u \rangle = \langle v \mid v \rangle$ would make $u + v$ isotropic, since then $\langle u + v \mid u + v \rangle = (1 + 1)\langle u \mid u \rangle = 0$. The equality of the two rays therefore forces the characteristic to differ from 2.

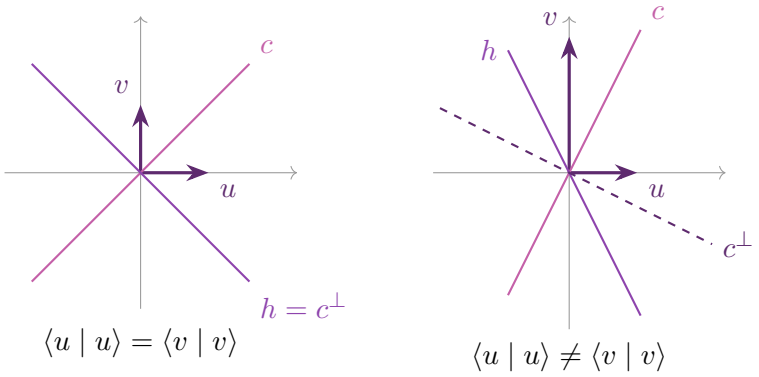


FIGURE 4. The ray $c = \Delta(u + v)$, its harmonic conjugate $h = \Delta(u - v)$, and its orthocomplement in the plane of u and v. They coincide when u and v have equal length (left), and differ otherwise (right).

Solèr's condition, in the form we use, asks for an infinite orthogonal sequence of atoms along which the orthocomplement and the harmonic conjugate agree at every step:

(S) There are mutually orthogonal atoms p_n, for $n \in \mathbb{N}$, and atoms $c_n \leq p_n \vee p_{n+1}$ with $c_n \neq p_n, p_{n+1}$, such that, for every $n \in \mathbb{N}$, the harmonic conjugate of c_n with respect to p_n and p_{n+1} is $c_n^\perp \wedge (p_n \vee p_{n+1})$.

This formulation of Solèr's condition through harmonic conjugates is due to Holland [Hol95, Theorems 3.7 and 4.1]; Morash's angle-bisecting axiom [Mor73], which asks for bisectors as in the left half of Figure 4, may be used in its place (see Holland [Hol95, §4]).

Lemma 3.4. *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), and let E be as in Theorem 3.3. After a rescaling of the form, the space E contains an infinite orthonormal sequence.*

Proof. Choose e_0 with $R_{p_0} = \Delta e_0$, and inductively choose e_{n+1} with $R_{p_{n+1}} = \Delta e_{n+1}$ and $R_{c_n} = \Delta(e_n + e_{n+1})$; this is possible because R_{c_n} is a ray of the plane spanned by e_n and e_{n+1}, distinct from Δe_n and Δe_{n+1}. By (1) and the computation that follows it, condition (S) gives $\langle e_n \mid e_n \rangle = \langle e_{n+1} \mid e_{n+1} \rangle$ for every $n \in \mathbb{N}$; call this common value ζ, which satisfies $\zeta^* = \zeta \neq 0$. The form $\langle x \mid y \rangle' = \langle x \mid y \rangle \zeta^{-1}$ is hermitian for the involution $\alpha \mapsto \zeta \alpha^* \zeta^{-1}$, has the same orthogonality relation, and satisfies $\langle e_n \mid e_n \rangle' = 1$ for every n. ☺

Definition 3.5. A *Solèr lattice* is an ortholattice that satisfies (C), (OM), (A), (CP), (I), (D) and (S). A *pre-Solèr lattice* is an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), but not necessarily (C).

Theorem 3.6 (Piron, Amemiya and Araki, Solèr; in this form Holland [Hol95, Theorem 4.1]**).** *An ortholattice is a Solèr lattice if and only if it is isomorphic to $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

Sketch. Let Λ be a Solèr lattice, with γ and E as in Theorem 3.3. Given $M \in \mathcal{C}(E)$, completeness provides the join a of the atoms p with $R_p \subseteq M$, and $\gamma(a)^\perp = \gamma(a^\perp) = M^\perp$, whence $\gamma(a) = M$; thus γ is an isomorphism, $\mathcal{C}(E)$ is orthomodular, and E is an orthomodular space by Proposition 2.8. By Lemma 3.4, the space E contains, after a rescaling, an infinite orthonormal sequence, and Solèr's theorem [Sol95] (see also Prestel [Pre95]) shows that E is a Hilbert space over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Conversely, $L(H)$ is complete and satisfies the other conditions; condition (S) holds with p_n the ray of e_n and c_n the ray of $e_n + e_{n+1}$, for an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$. ☺

Keller [Kel80] constructed orthomodular hermitian spaces that are not Hilbert spaces; hence condition (S) cannot be omitted from Theorem 3.6.

4. What a pre-Solèr lattice is missing (optional)

A pre-Solèr lattice satisfies every condition that defines a Solèr lattice except (C). We determine what it lacks: some elements, which an order-theoretic completion supplies, and possibly the orthomodularity of that completion. Neither deficiency can be detected by a sentence of $L_{\omega_1,\omega}$ about the lattice alone (see Proposition 4.5 below); this is why the vocabulary of Section 5 has vector sorts. From now on, Λ is a pre-Solèr lattice; we fix a hermitian space E that Theorem 3.3 provides for Λ, and we identify Λ with its image $\gamma(\Lambda) \subseteq \mathcal{C}(E)$.

Theorem 4.1. *Let Λ be a pre-Solèr lattice, identified as above with its image in $\mathcal{C}(E)$. Then Λ is join-dense and meet-dense in $\mathcal{C}(E)$; consequently $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is isomorphic as an ortholattice to the MacNeille completion of Λ.*

Proof. Every $M \in \mathcal{C}(E)$ is the join in $\mathcal{C}(E)$ of the rays it contains, which belong to Λ by Theorem 3.3. If $M^\perp = \{0\}$, then M is E, which belongs to Λ; otherwise M equals $M^{\perp\perp} =$

$\bigcap\{y^\perp : 0 \neq y \in M^\perp\}$, a meet of hyperplanes, each of which belongs to Λ. Proposition 1.5 gives both assertions. ☺

The elements of the MacNeille completion that do not come from Λ are therefore orthogonally closed subspaces M with $\dim M = \dim M^\perp = \infty$, since Λ contains every finite-dimensional subspace and the orthogonal complement of each. Adjoining them need not produce some $L(H)$, because the completion may fail to be orthomodular (see Example 4.3); the next theorem shows that orthomodularity of the completion is exactly what is needed.

Theorem 4.2. *For a pre-Solèr lattice Λ, the following are equivalent:*

(1) the MacNeille completion of Λ is orthomodular;
(2) the hermitian space E that Theorem 3.3 provides for Λ is orthomodular;
(3) Λ is isomorphic to a join-dense and meet-dense sub-ortholattice of $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

When these conditions hold, E with its form suitably rescaled is a Hilbert space as in (3), and Λ contains every finite-dimensional subspace of E and the orthogonal complement of each. Moreover, Λ is isomorphic to $L(H)$ for such an H if and only if Λ is complete.

Proof. Conditions (1) and (2) are equivalent by Theorem 4.1 and Proposition 2.8. If (2) holds, Lemma 3.4 and Solèr's theorem show that E, with its form rescaled, is a Hilbert space; rescaling changes neither $\mathcal{C}(E)$ nor its orthocomplementation, and Theorem 4.1 gives (3). If (3) holds, Proposition 1.5 identifies $L(H)$, which is orthomodular, with the MacNeille completion of Λ, together with its orthocomplementation; this is (1). For the last assertion, a complete lattice coincides with its MacNeille completion, and $L(H)$ is complete. ☺

Example 4.3. Let E_0 be the space of finitely supported complex sequences, with the inner product $\langle x \mid y\rangle = \sum_n x_n\overline{y_n}$ of the space ℓ^2 of square-summable sequences, and let Λ_0 be the set of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Then Λ_0 is a sub-ortholattice of $\mathcal{C}(E_0)$, since $F \vee G^\perp = (F^\perp \cap G)^\perp$ for finite-dimensional F and G. It is orthomodular, because every $M \in \Lambda_0$ satisfies $M + M^\perp = E_0$ by Lemma 2.6, and the argument of Proposition 2.8 applies. Since it contains every ray of E_0, it satisfies (A), (CP) and (I); conditions (D) and (S) hold with $p_n = \mathbb{C}e_n$ and $c_n = \mathbb{C}(e_n + e_{n+1})$, where $(e_n)_{n\in\mathbb{N}}$ is the sequence of standard unit vectors. Its MacNeille completion, $\mathcal{C}(E_0)$ by Theorem 4.1, is not orthomodular: by the Amemiya–Araki theorem [AA66], a complex inner product space whose lattice of orthogonally closed subspaces is orthomodular is complete, and E_0 is not. The lattice Λ_0 is thus an orthomodular lattice whose MacNeille completion is not orthomodular; Harding [Har91] studies such lattices.

Orthomodularity of the completion can be stated through the atoms of Λ alone. For a set $\mathcal{A}$ of atoms, let $\mathcal{A}^\perp$ be the set of atoms orthogonal to every element of $\mathcal{A}$, and call $\mathcal{A}$ *orthoclosed* when $\mathcal{A} = \mathcal{A}^{\perp\perp}$. Consider the following condition, named (MOM) for MacNeille orthomodularity:

(MOM) for every orthoclosed set $\mathcal{A}$ of atoms and every atom $p \notin \mathcal{A}$, some atom $q \in \mathcal{A}^\perp$ lies in $(\mathcal{A} \cup \{p\})^{\perp\perp}$.

Theorem 4.4. *The MacNeille completion of a pre-Solèr lattice is orthomodular if and only if the lattice satisfies (MOM).*

Proof. The map $\mathcal{A} \mapsto \bigcup\{R_p : p \in \mathcal{A}\}$ is a bijection from the orthoclosed sets of atoms onto $\mathcal{C}(E)$, carrying $\mathcal{A}^\perp$ to the orthogonal complement of the image of $\mathcal{A}$. If M is the image of $\mathcal{A}$ and $R_p = \Delta x$, then by Lemma 2.7 the set $(\mathcal{A} \cup \{p\})^{\perp\perp}$ corresponds to $M + \Delta x$. Condition (MOM) therefore says that $(M + \Delta x) \cap M^\perp \neq \{0\}$ whenever $M \in \mathcal{C}(E)$ and $x \notin M$. If

$y = v + \alpha x$ is nonzero, with $v \in M$ and $y \in M^\perp$, then the scalar α is nonzero, since $y \notin M$, and $x = \alpha^{-1}(y - v)$ lies in $M + M^\perp$; conversely, if $x = v + y$ with $v \in M$ and $y \in M^\perp$, then $y = x - v$ is a nonzero element of $(M + \Delta x) \cap M^\perp$. Thus (MOM) is equivalent to the orthomodularity of E, and Theorem 4.2 applies. ☺

When $\mathcal{A}$ corresponds to a finite-dimensional subspace, the requirement in (MOM) holds automatically, by Lemma 2.6; its content lies in the infinite orthoclosed sets of atoms, over which it quantifies.

Proposition 4.5. *Neither (C) nor (MOM) is equivalent, among pre-Solèr lattices, to a sentence of two-valued $L_{\omega_1,\omega}$ in the vocabulary of ortholattices expanded by constants for the atoms p_n and c_n of (S).*

Proof. Suppose that a sentence φ of $L_{\omega_1,\omega}$ were equivalent to (MOM) among pre-Solèr lattices. The axioms of ortholattices and the conditions (OM), (A), (CP), (I), (D) and (S) are expressed by countably many sentences of $L_{\omega_1,\omega}$ in this vocabulary; together with φ, they hold in $L(\ell^2)$. By the downward Löwenheim–Skolem theorem for a countable fragment, they hold in a countable structure Λ', which is a pre-Solèr lattice satisfying (MOM). By Theorems 4.4 and 4.2, the lattice Λ' contains every ray of an infinite-dimensional Hilbert space, and is therefore uncountable, a contradiction. The same argument applies to (C), using Theorem 3.6. ☺

5. The logic and the vocabulary

Completeness, the one property that separates the lattices $L(H)$ from the pre-Solèr lattices, cannot be expressed by sentences of $L_{\omega_1,\omega}$, two-valued or real-valued, about the lattice alone: by the downward Löwenheim–Skolem theorem, any countable set of such sentences that holds in some $L(H)$ also holds in a countable lattice, while every Solèr lattice is uncountable (for the details, see the optional Proposition 4.5). A Hilbert space can supply the missing joins: we place the lattice beside a Hilbert space and a space of sequences of vectors, and connect the three by symbols whose intended meanings are projection, membership and closed span.

5.1. **Real-valued $L_{\omega_1,\omega}$.** We use the real-valued infinitary logic of the first author [Aub], which extends both Eagle's infinitary $[0,1]$-valued logic [Eag14] and Keisler's real-valued logic [Kei23]; Keisler's model existence theorem [Kei71] extends to it. Of this logic we use only the following features:

- ♠ The vocabulary is countable and many-sorted; no metric is given a priori, and function symbols need not be continuous.
- ♠ Predicate symbols are real-valued, each with values in a fixed bounded interval; every formula therefore takes its values in a bounded interval, whose bound depends on the formula.
- ♠ Formulas are closed under continuous connectives, under the quantifiers sup and inf over each sort, and under countable suprema $\sup_n \varphi_n$ and infima $\inf_n \varphi_n$ of families of formulas that are uniformly bounded and have finitely many free variables in all.
- ♠ There is no equality symbol; as in [Kei23], a structure is *reduced* when any two elements that agree in every atomic formula with parameters are equal. Models are taken to be reduced.

A *condition* is an expression $\varphi = 0$ or $\varphi \leq 0$, where φ is a formula; an axiom is a condition required to hold for all values of its free variables. Implications between conditions are

expressed by the following device: if φ takes only nonnegative values and $|\psi| \le r_\psi$, then the condition

$$\psi \le \sup_{n\ge 1} \min(r_\psi, n\varphi) \tag{2}$$

says that $\varphi = 0$ implies $\psi \le 0$, since the right side is 0 when $\varphi = 0$ and r_ψ otherwise. The family $\min(r_\psi, n\varphi)$, for $n \ge 1$, is uniformly bounded, which is all that a countable supremum requires.

5.2. **The vocabulary.** The vocabulary $\mathcal{L}$ has three sorts. The two vector sorts are intended to be the closed unit balls of Hilbert spaces; this keeps every predicate bounded.

- The *Hilbert sort* $\mathbf{H}$ has, for each pair λ, λ' in the field $\mathbb{Q}(i)$ of complex numbers with rational real and imaginary parts, with $|\lambda| + |\lambda'| \le 1$, a binary function $(x, y) \mapsto \lambda x + \lambda' y$; the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$, with values in $[-1, 1]$; and constants e_n, for $n \in \mathbb{N}$. We write 0 for the term $0x + 0x$, and $\langle x \mid y\rangle$ for the pair of predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$.
- The *sequence sort* $\mathbf{K}$ has the same functions and predicates as $\mathbf{H}$, and one constant ε.
- The *Solèr sort* $\mathbf{L}$ has the constants 0 and 1, the binary functions $\wedge$ and $\vee$, the unary function $^\perp$, and a predicate $d(a, b)$ with values in $[0, 1]$, the *discrete predicate*.
- Function symbols connecting the sorts:

$$\pi\colon \mathbf{L} \times \mathbf{H} \to \mathbf{H}, \quad \mathrm{at}\colon \mathbf{H} \to \mathbf{L}, \quad \mathrm{rep}\colon \mathbf{L} \to \mathbf{H},$$
$$\iota_n\colon \mathbf{H} \to \mathbf{K}, \quad \kappa_n\colon \mathbf{K} \to \mathbf{H}\ (n \in \mathbb{N}), \quad \mathrm{span}\colon \mathbf{K} \to \mathbf{L}.$$

- A predicate μ on $\mathbf{L}$, the *measure*, with values in $[-1, 2]$; the interval is wider than $[0, 1]$ so that axiom (M1), not the vocabulary, confines μ to $[0, 1]$.

For vector terms t_1, t_2 of the same sort, the equation $t_1 \doteq t_2$ abbreviates the condition $\|\frac{1}{2}t_1 - \frac{1}{2}t_2\| = 0$; for terms of the Solèr sort, $t_1 = t_2$ abbreviates $d(t_1, t_2) = 0$. A vector x *belongs* to an element a of the Solèr sort, or is a *member* of a, when $\pi(a, x) \doteq x$. The intended meanings are as follows: $\pi(a, \cdot)$ is the orthogonal projection onto the closed subspace spanned by the members of a; for $x \ne 0$, $\mathrm{at}(x)$ is the element whose members are the multiples of x in the unit ball; $\mathrm{rep}(a)$ is a unit vector belonging to a when $a \ne 0$; (e_n) is an orthonormal basis; ι_n places a vector in position n of a sequence and κ_n reads position n; ε is the sequence $(2^{-n-1}e_n)_n$; and $\mathrm{span}(s)$ is the element whose members are the vectors of the unit ball in the closed span of the entries of s.

Conditions on the values of the measure can be as complicated as Borel sets. For a Borel set $B \subseteq \mathbb{R}$, we write $\mathbf{1}_B$ for its indicator function, which is 1 on B and 0 off B.

Lemma 5.1. *For every Borel set $B \subseteq \mathbb{R}$, there is a formula $\mathbf{1}_B(\mu(a))$ of real-valued $L_{\omega_1,\omega}$, with the single free variable a of the Solèr sort, whose value at every element a of every structure is $\mathbf{1}_B(\mu(a))$.*

Proof. Let $\mathcal{B}$ be the class of sets $B \subseteq \mathbb{R}$ for which such a formula exists. For rational numbers $\alpha \le \beta$, the distance from a real number t to the interval $[\alpha, \beta]$ is $\max(0, \alpha - t, t - \beta)$, a continuous function of t; hence

$$\inf_{n\ge 1} \max\bigl(0,\ 1 - n\max(0, \alpha - \mu(a), \mu(a) - \beta)\bigr)$$

is a formula, and its value is 1 when $\mu(a) \in [\alpha, \beta]$ and 0 otherwise. Thus $\mathcal{B}$ contains the closed intervals with rational endpoints. If $B \in \mathcal{B}$, then $1 - \mathbf{1}_B(\mu(a))$ is a formula for $\mathbb{R} \setminus B$; if $B_n \in \mathcal{B}$ for every $n \in \mathbb{N}$, then the family of formulas $\mathbf{1}_{B_n}(\mu(a))$ is bounded by 1, and

its supremum is a formula for $\bigcup_n B_n$. Hence $\mathcal{B}$ is a σ-algebra. Every open subset of $\mathbb{R}$ is a countable union of closed intervals with rational endpoints, so $\mathcal{B}$ contains every Borel set. ☺

The nesting of countable suprema in the formula $\mathbf{1}_B(\mu(a))$ follows the construction of B from intervals, and may have any countable depth.

Definition 5.2. A structure $\mathcal{M}$ for $\mathcal{L}$ is *standard* when there are complex Hilbert spaces H and K such that the Hilbert sort and the sequence sort of $\mathcal{M}$ are the closed unit balls of H and K, and the functions $\lambda x + \lambda' y$ and the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$ are the restrictions of the operations, the inner product and the norm of H and of K.

Standardness is a restriction on the structures we consider, not a condition imposed by an axiom. Since $\mathcal{L}$ is countable and the theories T_0, T_1 and T of Section 6 below are countable, the downward Löwenheim–Skolem theorem gives them models whose vector sorts are countable, hence not complete; Theorem C applies to such models, whereas the conclusions of Theorem A(1) and Theorem B fail in them (see Remark 11.2).

6. The axioms

The axioms come in groups, one for each sort, one for each family of symbols connecting the sorts, and one for the measure. Together they say that the elements of the Solèr sort name orthogonal projections of the Hilbert sort, ordered and combined as closed subspaces are; that atoms correspond to unit vectors up to a phase; that every sequence of vectors names its closed span; and that μ is σ-additive. Free variables are universally quantified: x, y, z range over $\mathbf{H}$, s, t over $\mathbf{K}$, and a, b, c over $\mathbf{L}$. The letters λ, λ' (ranging over $\mathbb{Q}(i)$, with $|\lambda| + |\lambda'| \leq 1$) and n, m, k, N (ranging over $\mathbb{N}$) are not variables of the logic: an axiom containing them stands for countably many axioms, one for each value. Readings in words follow the axioms where useful.

(H) The Hilbert sort.

(H1) *Inner product space.* For any two terms built from variables by the functions $\lambda x+\lambda' y$ that denote the same $\mathbb{Q}(i)$-linear combination of the variables, the equation between them; the conditions $\langle \lambda x + \lambda' y \mid z\rangle = \lambda\langle x \mid z\rangle + \lambda'\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \overline{\langle x \mid y\rangle}$, each written as two real conditions; and $\|x\| = \sqrt{\max(0, \mathrm{Re}\langle x \mid x\rangle)}$.

(H2) *Separability.* $\langle e_n \mid e_m\rangle = \delta_{nm}$, where δ_{nm} is the Kronecker delta, and

$$\|x\|^2 = \sup_N \min\Big(1, \sum_{n<N} |\langle x \mid e_n\rangle|^2\Big).$$

The constants e_n are orthonormal and satisfy Parseval's identity.

(OL) The Solèr sort.

(OL1) The equations of ortholattices:

$$a \wedge b = b \wedge a, \quad a \wedge (b \wedge c) = (a \wedge b) \wedge c, \quad a \wedge (a \vee b) = a,$$
$$a \vee b = b \vee a, \quad a \vee (b \vee c) = (a \vee b) \vee c, \quad a \vee (a \wedge b) = a,$$
$$a \wedge 0 = 0, \quad a \vee 1 = 1, \quad a^{\perp\perp} = a, \quad (a \wedge b)^{\perp} = a^{\perp} \vee b^{\perp}, \quad a \wedge a^{\perp} = 0.$$

(OL2) *The discrete predicate is equality.* $\min(d(a,b), 1-d(a,b)) = 0$; $d(a,a) = 0$; $|d(a,c)-d(b,c)| \leq d(a,b)$; and $d(a \wedge c, b \wedge c) \leq d(a,b)$, $d(a \vee c, b \vee c) \leq d(a,b)$, $d(a^{\perp}, b^{\perp}) \leq$

$d(a,b)$, $\|\frac{1}{2}\pi(a,x) - \frac{1}{2}\pi(b,x)\| \le d(a,b)$, $\|\frac{1}{2}\mathrm{rep}(a) - \frac{1}{2}\mathrm{rep}(b)\| \le d(a,b)$ and $|\mu(a) - \mu(b)| \le 3\,d(a,b)$.

(P) Each element of the Solèr sort determines a projection.

(P1) $\pi(a, \lambda x + \lambda' y) \doteq \lambda\pi(a,x) + \lambda'\pi(a,y)$.
(P2) $\pi(a, \pi(a,x)) \doteq \pi(a,x)$.
(P3) $\langle \pi(a,x) \mid y\rangle = \langle x \mid \pi(a,y)\rangle$.
(P4) *Extensionality.* $d(a,b) \le \sup_{n\ge 1} \min\Big(1,\, n \sup_x \big\|\frac{1}{2}\pi(a,x) - \frac{1}{2}\pi(b,x)\big\|\Big)$.
Two elements with the same projection are equal.

(Sub) The order and the operations of the Solèr sort are those of subspaces.

(Sub0) $\pi(0,x) \doteq 0$ and $\pi(1,x) \doteq x$.
Only the zero vector belongs to 0*; every vector belongs to* 1.
(Sub1) $\|\frac{1}{2}\pi(b,\pi(a,x)) - \frac{1}{2}\pi(a,x)\| \le d(a \wedge b, a)$, and $d(a \wedge b, a) \le \sup_{n\ge 1} \min\Big(1,\, n \sup_x \big\|\frac{1}{2}\pi(b,\pi(a,x)) - \frac{1}{2}\pi(a,x)\big\|\Big)$.
The relation $a \le b$ holds if and only if every member of a is a member of b.
(Sub2) $\inf_N \sup_{n\ge N} \big\|\frac{1}{2}\pi(a\wedge b, x) - \frac{1}{2}(P_a P_b)^n x\big\| = 0$, where $(P_aP_b)^n x$ is the term obtained by applying $\pi(b,\cdot)$ and then $\pi(a,\cdot)$, n times.
The members of $a \wedge b$ are the common members of a and b.
(Sub3) $\inf_N \sup_{n\ge N} \big\|\frac{1}{4}\pi(a\vee b, x) - \frac{1}{4}x + \frac{1}{4}(P_{a^\perp}P_{b^\perp})^n x\big\| = 0$, the vector inside the norm being denoted by a term built from the functions $\lambda x + \lambda' y$.
The closed subspace spanned by the members of $a \vee b$ is the closure of the sum of the subspaces spanned by the members of a and of b.
(Sub4) $\frac{1}{2}\pi(a^\perp, x) \doteq \frac{1}{2}x - \frac{1}{2}\pi(a,x)$.
The members of $a^\perp$ are the vectors orthogonal to all members of a.

Let $\chi(a)$ be the formula $\sup_b \min\big(d(b,0),\, d(b,a),\, 1 - d(a\wedge b, b)\big)$; it takes the value 1 if some b lies strictly between 0 and a, and 0 otherwise.

(At) Atoms and unit vectors.

(At1) $\pi(\mathrm{at}(x), x) \doteq x$, and $\big|\langle \pi(\mathrm{at}(x),y) \mid \pi(\mathrm{at}(x),z)\rangle\big|^2 = \|\pi(\mathrm{at}(x),y)\|^2\,\|\pi(\mathrm{at}(x),z)\|^2$.
Every vector x belongs to $\mathrm{at}(x)$*, and any two members of* $\mathrm{at}(x)$ *are linearly dependent.*
(At2) $\pi(a, \mathrm{rep}(a)) \doteq \mathrm{rep}(a)$, and $\big|\,\|\mathrm{rep}(a)\| - 1\,\big| \le 1 - d(a,0)$.
The vector $\mathrm{rep}(a)$ *belongs to a, and it is a unit vector when $a \ne 0$.*
(At3) $d(\mathrm{at}(\mathrm{rep}(a)), a) \le \max\big(\chi(a),\, 1 - d(a,0)\big)$.
Every atom a equals $\mathrm{at}(\mathrm{rep}(a))$.
(At4) $\|x\|\,\|y\| - |\langle x \mid y\rangle| \le d(\mathrm{at}(x), \mathrm{at}(y))$, and

$$d(\mathrm{at}(x),\mathrm{at}(y)) \le \sup_{n\ge 1} \min\Big(1,\, n\big(\|x\|\,\|y\| - |\langle x \mid y\rangle| + \big|1 - \|x\|\big| + \big|1 - \|y\|\big|\big)\Big).$$

Two unit vectors determine the same atom if and only if one is a multiple of the other by a complex number of modulus one.

(At5) $\mathrm{at}(0) = 0$, and $d(\mathrm{at}(x), \mathrm{at}(y)) \leq \sup_{n\geq 1} \min\big(1,\, n\|\tfrac{1}{2}x - \tfrac{1}{2}y\|\big)$.

(K) The sequence sort.

(K1) The axioms (H1), written for **K**.

(K2) $\iota_n(\lambda x + \lambda' y) \doteq \lambda\iota_n x + \lambda'\iota_n y$ and $\kappa_n(\lambda s + \lambda' t) \doteq \lambda\kappa_n s + \lambda'\kappa_n t$; $\langle \iota_n x \mid \iota_m y\rangle = \delta_{nm}\langle x \mid y\rangle$; and $\langle \kappa_n s \mid x\rangle = \langle s \mid \iota_n x\rangle$.

(K3) $\|s\|^2 = \sup_N \min\Big(1, \sum_{n<N} \|\kappa_n s\|^2\Big)$.

A sequence is determined by its entries, and its norm is the ℓ^2 norm of its entries.

(K4) $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$.

(Sp) Spans.

(Sp1) $\pi(\mathrm{span}(s), \kappa_n s) \doteq \kappa_n s$.

Every entry of s belongs to $\mathrm{span}(s)$.

(Sp2) $\|\pi(\mathrm{span}(s), x)\|^2 \leq \sup_{m\geq 1} \min\Big(1,\, m \sup_N \min\Big(1, \sum_{n<N} |\langle x \mid \kappa_n s\rangle|^2\Big)\Big)$.

A vector orthogonal to every entry of s is orthogonal to every member of $\mathrm{span}(s)$.

(Sp3) $d(\mathrm{span}(s), \mathrm{span}(t)) \leq \sup_{n\geq 1} \min\big(1,\, n\|\tfrac{1}{2}s - \tfrac{1}{2}t\|\big)$.

Let $\theta(s)$ be the formula $\sup_N \min\big(1, \sum_{n,m<N,\, n\neq m} |\langle \kappa_n s \mid \kappa_m s\rangle|\big)$, which takes values in $[0,1]$ and vanishes exactly when the entries of s are mutually orthogonal.

(M) The measure.

(M1) $0 \leq \mu(a) \leq 1$.

(M2) $\mu(1) = 1$.

(M3) *σ-additivity.*

$$\Big|\, \mu(\mathrm{span}(s)) - \sup_N \min\Big(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n s))\Big)\Big| \leq \sup_{k\geq 1} \min\big(2,\, k\,\theta(s)\big).$$

If the entries of s are mutually orthogonal, the value of μ at $\mathrm{span}(s)$ *is the sum of its values at the elements* $\mathrm{at}(\kappa_n s)$.

Definition 6.1. The theory T_0 consists of the axioms (H), (OL), (P), (Sub) and (At), without the clause on μ in (OL2), in the vocabulary $\mathcal{L}_0$ obtained from $\mathcal{L}$ by removing the sequence sort with its constant ε, the functions ι_n, κ_n and span, and the predicate μ. The theory T_1 consists of T_0 together with (K) and (Sp), in the vocabulary $\mathcal{L}$ without μ; the theory T consists of T_1 together with (M) and the clause on μ in (OL2).

Lemma 6.2. *In a model of T_0, two elements a, b of the Solèr sort are equal if and only if $d(a,b) = 0$, and two vectors x, y are equal if and only if $\|\tfrac{1}{2}x - \tfrac{1}{2}y\| = 0$; the same holds in the sequence sort of a model of T_1.*

Proof. By (H1), (K1), (OL2), (P1)–(P3), (At5), (K2) and (Sp3), and by the Cauchy–Schwarz inequality for the predicates, every atomic formula takes the same values at a and b when $d(a,b)=0$, and at x and y when $\|\frac{1}{2}x-\frac{1}{2}y\|=0$; since models are reduced, the elements in question are equal. ☺

Lemma 6.3. *Let $\mathcal{M}$ be a model of T_0. There is a complex Hilbert space $\bar{E}$ such that the Hilbert sort of $\mathcal{M}$ can be identified with a subset B of the closed unit ball of $\bar{E}$ whose $\mathbb{Q}(i)$-linear span E is dense, the functions $\lambda x+\lambda' y$ and the predicates being the restrictions of those of $\bar{E}$. For every element a of the Solèr sort, $\pi(a,\cdot)$ extends uniquely to an orthogonal projection $\bar{P}_a$ of $\bar{E}$. If $\mathcal{M}$ is standard, then B is the whole closed unit ball of $\bar{E}$.*

Proof. On pairs (k,x), with $k\geq 1$ an integer and $x\in B$, read as kx, put $(k,x)\sim(l,y)$ when $\|\frac{1}{2l}x-\frac{1}{2k}y\|=0$; define the sum by $(k,x)+(l,y)=(2kl,\frac{1}{2l}x+\frac{1}{2k}y)$ and the form by $\langle(k,x)\mid(l,y)\rangle=kl\langle x\mid y\rangle$; multiplication by elements of $\mathbb{Q}(i)$ is defined similarly. Axiom (H1) makes the quotient a $\mathbb{Q}(i)$-vector space E with a positive definite hermitian form, and Lemma 6.2 makes the map from B to E injective; we let $\bar{E}$ be the completion of E. By (P1)–(P3), the map $\pi(a,\cdot)$ extends to a $\mathbb{Q}(i)$-linear idempotent map Q_a on E, symmetric with respect to the form; hence $\|Q_a x\|^2=\langle x\mid Q_a x\rangle\leq\|x\|\|Q_a x\|$, the map Q_a is contractive, and it extends by continuity to a self-adjoint idempotent $\bar{P}_a$ of $\bar{E}$. In a standard model, the set B is the closed unit ball of a Hilbert space, which is complete and spanned by B; that space is therefore $\bar{E}$. ☺

7. Standard models of T_0: pre-Solèr lattices

In a standard model of T_0, the theory without the sequence sort and without the measure, the Solèr sort is a pre-Solèr lattice whose image is dense, in the order-theoretic sense of Section 1, in the lattice of closed subspaces of the Hilbert space; and the σ-additive states on it, defined below, already come from density operators. Throughout this section, $\mathcal{M}$ is a standard model of T_0. We write H for the Hilbert space whose closed unit ball is the Hilbert sort of $\mathcal{M}$ (see Definition 5.2), $\mathbf{L}^{\mathcal{M}}$ for its Solèr sort, P_a for the projection $\bar{P}_a$ of Lemma 6.3, and S_a for the range of P_a, the closed subspace spanned by the members of a.

Proposition 7.1. *The space H is a separable infinite-dimensional complex Hilbert space, and $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H.*

Proof. By (H2) the vectors e_n are orthonormal, and Parseval's identity shows that every vector of the unit ball, and hence of H, lies in their closed span; thus no nonzero vector is orthogonal to all of them. ☺

Theorem 7.2 (von Neumann's alternating projection theorem [vN49, Lemma 22]**).** *Let U and V be closed subspaces of a Hilbert space, and let P_U, P_V and $P_{U\cap V}$ be the orthogonal projections onto U, V and $U\cap V$. Then $(P_U P_V)^n x$ converges in norm to $P_{U\cap V}x$ for every vector x.*

Proposition 7.3. *The map $\tau\colon a\mapsto S_a$ is an embedding of ortholattices from $\mathbf{L}^{\mathcal{M}}$ into $L(H)$; that is, it is injective and satisfies*

$$S_0=\{0\},\quad S_1=H,\quad S_{a\wedge b}=S_a\cap S_b,\quad S_{a\vee b}=\overline{S_a+S_b},\quad S_{a^\perp}=S_a^\perp.$$

Moreover, the relation $a\leq b$ holds if and only if $S_a\subseteq S_b$.

Proof. Axiom (Sub0) gives the first two equalities, and axiom (Sub4) gives $P_{a^\perp}=\mathrm{Id}-P_a$, whose range is $S_a^\perp$. Combined with Theorem 7.2, axiom (Sub2) shows that $P_{a\wedge b}=P_{S_a\cap S_b}$, and axiom (Sub3) that $P_{a\vee b}=\mathrm{Id}-P_{S_a^\perp\cap S_b^\perp}$, whose range is $(S_a^\perp\cap S_b^\perp)^\perp=\overline{S_a+S_b}$. If $S_a=S_b$,

then $P_a = P_b$, and axiom (P4) gives $a = b$. Finally, axiom (Sub1) shows that $a \wedge b = a$ exactly when $P_b P_a = P_a$, that is, when $S_a \subseteq S_b$. ☺

Proposition 7.4. *For every nonzero x in the unit ball, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}x$. The atoms of $\mathbf{L}^{\mathcal{M}}$ are exactly the elements $\mathrm{at}(x)$ with $x \neq 0$, and $x \mapsto \mathrm{at}(x)$ induces a bijection from the unit sphere of H, modulo multiplication by complex numbers of modulus one, onto the set of atoms of $\mathbf{L}^{\mathcal{M}}$, whose inverse sends an atom a to the class of $\mathrm{rep}(a)$.*

Proof. By (At1), the vector x lies in $S_{\mathrm{at}(x)}$, and any two vectors of $S_{\mathrm{at}(x)}$ attain equality in the Cauchy–Schwarz inequality, hence are linearly dependent; therefore $S_{\mathrm{at}(x)} = \mathbb{C}x$. Since this ray is an atom of $L(H)$ and τ is an order embedding, $\mathrm{at}(x)$ is an atom of $\mathbf{L}^{\mathcal{M}}$. If a is an atom of $\mathbf{L}^{\mathcal{M}}$, then axiom (At3) gives $a = \mathrm{at}(\mathrm{rep}(a))$. For unit vectors x and y, axiom (At4) gives $\mathrm{at}(x) = \mathrm{at}(y)$ if and only if $|\langle x \mid y\rangle| = 1$, that is, if and only if y is a multiple of x by a complex number of modulus one. ☺

Theorem 7.5. *Let $\mathcal{M}$ be a standard model of T_0. Its Solèr sort $\mathbf{L}^{\mathcal{M}}$ is a pre-Solèr lattice, and the image of the map $\tau\colon a \mapsto S_a$ is join-dense and meet-dense in $L(H)$, which is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. The image of τ contains every finite-dimensional subspace of H and the orthogonal complement of each, and τ maps every join that exists in $\mathbf{L}^{\mathcal{M}}$ to the corresponding join in $L(H)$.*

Proof. If F is a finite-dimensional subspace with a basis $x_1, \ldots, x_k$ of unit vectors, then $\tau(\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k)) = \overline{F} = F$ by Propositions 7.3 and 7.4, and the orthogonal complement $F^\perp$ is $\tau\big((\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^\perp\big)$. The lattice $\mathbf{L}^{\mathcal{M}}$ is orthomodular because $\tau(\mathbf{L}^{\mathcal{M}})$ is a subortholattice of the orthomodular lattice $L(H)$. If $a \neq 0$, the atoms below a are the elements $\mathrm{at}(x)$ with $x \in S_a$ and $0 < \|x\| \leq 1$; any upper bound b of them in $\mathbf{L}^{\mathcal{M}}$ satisfies $S_b \supseteq S_a$, hence $b \geq a$, and a is their join; thus $\mathbf{L}^{\mathcal{M}}$ is atomistic. For the covering property, let $\mathrm{at}(x)$ be an atom not below a; then $x \notin S_a$, and Proposition 7.3 and Lemma 2.7 give $\tau(a \vee \mathrm{at}(x)) = S_a + \mathbb{C}x$, in which S_a has codimension one; since τ is an order embedding, no element lies strictly between a and $a \vee \mathrm{at}(x)$. The join of two distinct atoms corresponds to a plane, which contains infinitely many rays, all in $\tau(\mathbf{L}^{\mathcal{M}})$; this gives (I). The atoms $\mathrm{at}(e_n)$ give (D), and condition (S) holds with $p_n = \mathrm{at}(e_n)$ and $c_n = \mathrm{at}(\frac{1}{2}(e_n + e_{n+1}))$, by (1) and the computation following it.

Every closed subspace S of H is the join in $L(H)$ of its rays, and it equals $\bigcap\{y^\perp : y \in S^\perp,\ 0 < \|y\| \leq 1\}$, where $y^\perp = S_{\mathrm{at}(y)^\perp}$ lies in the image of τ; these two facts are join-density and meet-density, and Proposition 1.5 identifies $L(H)$ with the MacNeille completion of $\mathbf{L}^{\mathcal{M}}$. Finally, let $A \subseteq \mathbf{L}^{\mathcal{M}}$ have a join c in $\mathbf{L}^{\mathcal{M}}$, and let W be the closed span of $\bigcup_{a \in A} S_a$; clearly $W \subseteq S_c$. If y is a vector of $S_c \cap W^\perp$ with $0 < \|y\| \leq 1$, then every $a \in A$ lies below $\mathrm{at}(y)^\perp$, hence so does c, and y is orthogonal to itself, which is absurd. Since $S_c = W + (S_c \cap W^\perp)$, it follows that $S_c = W$. ☺

Example 7.6. Let L_{fc} (for "finite dimension or finite codimension") be the set of finite-dimensional subspaces of ℓ^2 and their orthogonal complements. With d the discrete metric, $\pi(S, x) = P_S x$, $\mathrm{at}(x) = \mathbb{C}x$, a unit vector $\mathrm{rep}(S) \in S$ chosen for each $S \neq \{0\}$, $\mathrm{rep}(\{0\}) = 0$, and the standard unit vectors as (e_n), the lattice L_{fc} is the Solèr sort of a standard model of T_0, although it is not complete. The axioms of T_0 constrain only the elements that the Solèr sort contains; none of them demands an element whose members span a given closed subspace of infinite dimension and infinite codimension. The sequence sort, which the axioms of T_1 govern, is introduced to make that demand.

A *density operator* on H is a positive operator ρ whose trace $\operatorname{Tr}\rho = \sum_k \langle \rho u_k \mid u_k \rangle$, computed in any orthonormal basis $(u_k)_{k\in\mathbb{N}}$ of H, equals 1; quantum mechanics represents the states of a system by density operators. A *σ-additive state* on an ortholattice Λ is a map $\nu\colon \Lambda \to [0,1]$ with $\nu(1) = 1$ such that $\nu(\bigvee_n a_n) = \sum_n \nu(a_n)$ for every finite or countable family of mutually orthogonal elements whose join exists in Λ.

Theorem 7.7 (Gleason's theorem for T_0). *Let $\mathcal{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathcal{M}}$, let P_a be the orthogonal projection onto the closed subspace spanned by the members of a. The σ-additive states on $\mathbf{L}^{\mathcal{M}}$ are exactly the maps $a \mapsto \operatorname{Tr}(\rho P_a)$, where ρ ranges over the density operators on H, and each such state determines its density operator.*

Proof. If the elements a_n, for n in a finite or countable set, are mutually orthogonal with join c in $\mathbf{L}^{\mathcal{M}}$, then by Theorem 7.5 the subspace S_c is the closure of $\sum_n S_{a_n}$; hence $P_c x = \sum_n P_{a_n} x$ for every $x \in H$, and $\operatorname{Tr}(\rho P_c) = \sum_n \operatorname{Tr}(\rho P_{a_n})$; moreover $\operatorname{Tr}(\rho P_1) = \operatorname{Tr}\rho = 1$. Conversely, let ν be a σ-additive state; additivity on the pair $1, 0$ gives $\nu(0) = 0$. For a unit vector x, put $f(x) = \nu(\operatorname{at}(x))$. For $a \neq 0$ and an orthonormal basis (u_k) of S_a, which is countable, the atoms $\operatorname{at}(u_k)$ are mutually orthogonal and their join in $\mathbf{L}^{\mathcal{M}}$ is a, because every upper bound b of them satisfies $S_b \supseteq S_a$; hence $\nu(a) = \sum_k f(u_k)$. Taking $a = 1$ shows that f is a *frame function* of weight 1: its sum over every orthonormal basis of H is 1. Since H is separable of dimension at least three, Gleason's theorem [Gle57] gives a unique density operator ρ with $f(x) = \langle \rho x \mid x \rangle$ for every unit vector x, and then $\nu(a) = \sum_k \langle \rho u_k \mid u_k \rangle = \operatorname{Tr}(\rho P_a)$. ☺

8. Standard models of T_1 and the proofs of Theorems A and B

The axioms of T_0 cannot compel the Solèr sort to contain a given closed subspace of infinite dimension and infinite codimension, as the lattice L_{fc} of Example 7.6 shows. The sequence sort supplies what is missing: every closed subspace of a separable Hilbert space is the closed span of a square-summable sequence, and the function span names it. In this section, $\mathcal{M}$ is a standard model of T_1; we write H and K for the Hilbert spaces whose closed unit balls are its Hilbert sort and its sequence sort, and $\mathbf{L}^{\mathcal{M}}$ for its Solèr sort. Since T_1 contains T_0, the results of Section 7 apply to $\mathcal{M}$, and P_a, S_a and τ are as there.

Lemma 8.1. *The maps κ_n extend to bounded linear maps from K to H, and $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is a linear isometry of K onto the space $\ell^2(\mathbb{N}, H)$ of sequences $(x_n)_{n\in\mathbb{N}}$ in H with $\sum_n \|x_n\|^2 < \infty$.*

Proof. By (K2) and (K3), the map is linear and isometric, and $\kappa_m \iota_n x = \delta_{mn} x$. Given $(x_n) \in \ell^2(\mathbb{N}, H)$, the partial sums $\sum_{n<N} \iota_n x_n$ form a Cauchy sequence in K, by (K2); since K is complete, they converge to some s, and $\kappa_n s = x_n$ by the continuity of κ_n. ☺

Lemma 8.2. *For every s in the unit ball of K, the subspace $S_{\operatorname{span}(s)}$ is the closed span of the entries $\kappa_n s$.*

Proof. Let W be that closed span. By (Sp1), the subspace $S_{\operatorname{span}(s)}$ contains every entry of s; hence $W \subseteq S_{\operatorname{span}(s)}$. If $x \perp W$, axiom (Sp2) gives $P_{\operatorname{span}(s)} x = 0$; hence $S_{\operatorname{span}(s)} \subseteq W^{\perp\perp} = W$. ☺

Theorem 8.3. *Let $\mathcal{M}$ be a standard model of T_1, and let H be the Hilbert space whose closed unit ball is its Hilbert sort. Then H is a separable infinite-dimensional complex Hilbert space, and the map $\tau\colon a \mapsto S_a$ is an isomorphism of ortholattices from the Solèr sort of $\mathcal{M}$ onto $L(H)$; in particular, that Solèr sort is a Solèr lattice. Conversely, for every separable infinite-dimensional complex Hilbert space H', there is a standard model of T_1 whose Hilbert sort is the closed unit ball of H' and whose Solèr sort is $L(H')$.*

Proof. By Proposition 7.1, the space H is separable and infinite-dimensional. By Proposition 7.3, the map τ is an embedding; it remains to be shown that it is onto. Let S be a closed subspace of H, and let $(u_n)_{n\in\mathbb{N}}$ enumerate an orthonormal basis of S, which is countable by Proposition 7.1, followed by zeros if the basis is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_n s = 2^{-n-1}u_n$ for all n, since $\sum_n 4^{-n-1} < 1$, and Lemma 8.2 gives $S_{\mathrm{span}(s)} = S$.

For the converse, interpret the Solèr sort as $L(H')$, with d the discrete metric ($d(S,S') = 1$ when $S \neq S'$), $\pi(S,x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$; let $\mathrm{rep}(S)$ be a unit vector chosen in each $S \neq \{0\}$, and let $\mathrm{rep}(\{0\})$ be 0; let (e_n) be an orthonormal basis of H'; interpret the sequence sort as the unit ball of $\ell^2(\mathbb{N}, H')$, with $\iota_n x$ the sequence with x in position n and 0 elsewhere, $\kappa_n s$ the entry of s in position n, and $\varepsilon = (2^{-n-1}e_n)_n$; and let $\mathrm{span}(s)$ be the closed span of the entries of s. Every axiom holds by construction, axioms (Sub2) and (Sub3) by Theorem 7.2. ☺

Proof of Theorem A. Let $\mathcal{M}$ be a standard model of T; it is in particular a standard model of T_1, and by Theorem 8.3 the space H is a separable infinite-dimensional complex Hilbert space and τ is an isomorphism of $\mathbf{L}^{\mathcal{M}}$ onto $L(H)$.

Step 1: $\mu(0) = 0$. Let s be the zero element of the sequence sort. Its entries are 0; hence $\theta(s) = 0$, and Lemma 8.2 gives $S_{\mathrm{span}(s)} = \{0\}$, whence $\mathrm{span}(s) = 0$ by the injectivity of τ. Moreover, axiom (At5) gives $\mathrm{at}(\kappa_n s) = \mathrm{at}(0) = 0$. Axiom (M3) gives $\mu(0) = \sup_N \min(2, N\mu(0))$; if $\mu(0)$ were positive, the right side would be 2, contradicting (M1).

Step 2: sums along orthonormal bases. Let a be an element of $\mathbf{L}^{\mathcal{M}}$, and let $(u_k)_{k\in\mathbb{N}}$ enumerate an orthonormal basis of S_a, followed by zeros if it is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_k s = 2^{-k-1}u_k$. Its entries are mutually orthogonal, whence $\theta(s) = 0$; by Lemma 8.2 and the injectivity of τ, the element $\mathrm{span}(s)$ is a. For $u_k \neq 0$, Proposition 7.4 gives $S_{\mathrm{at}(2^{-k-1}u_k)} = \mathbb{C}u_k$, hence $\mathrm{at}(\kappa_k s) = \mathrm{at}(u_k)$; for $u_k = 0$, the element $\mathrm{at}(\kappa_k s)$ is 0, whose measure is 0 by Step 1. Axiom (M3) and $\mu(a) \leq 1$ give

$$\mu(a) = \sum_k \mu(\mathrm{at}(u_k)). \tag{$*$}$$

Step 3: application of Gleason's theorem. For a unit vector x, put $f(x) = \mu(\mathrm{at}(x))$. Applying $(*)$ to $a = 1$ and using (M2), we obtain $\sum_k f(u_k) = 1$ for every orthonormal basis (u_k) of H. Arguing as in the last three sentences of the proof of Theorem 7.7, with μ in place of ν and $(*)$ in place of the decomposition used there, we obtain a unique density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in \mathbf{L}^{\mathcal{M}}$.

The converse. Given a separable infinite-dimensional complex Hilbert space H' and a density operator ρ on H', expand the model of Theorem 8.3 for H' by $\mu(a) = \mathrm{Tr}(\rho P_a)$. The clause on μ in (OL2) holds because d is the discrete metric and $|\mu(a) - \mu(b)| \leq 1$. Axioms (M1) and (M2) hold because $\mathrm{Tr}(\rho P) \in [0,1]$ for every orthogonal projection P and $\mathrm{Tr}\,\rho = 1$. For (M3), suppose that the entries $x_n = \kappa_n s$ are mutually orthogonal. Then

$\operatorname{span}(s)$ is their closed span, and, with $\hat{x}_n = x_n/\|x_n\|$, we have $\operatorname{Tr}(\rho P_{\operatorname{span}(s)}) = \sum_{x_n \neq 0} \langle \rho \hat{x}_n \mid \hat{x}_n \rangle = \sum_n \mu(\operatorname{at}(x_n)) \leq 1$; the left side of (M3) is therefore 0. If the entries are not mutually orthogonal, the right side of (M3) is 2, while the left side is at most 2. ☺

The proof of Theorem B rests on the fact that a density operator, unlike a two-valued measure, takes every value in $[0,1]$ on closed subspaces.

Lemma 8.4. *Let H be a separable infinite-dimensional complex Hilbert space and ρ a density operator on H. Then $\{\operatorname{Tr}(\rho P_S) : S \in L(H)\} = [0,1]$.*

Proof. The values 0 and 1 are attained at $\{0\}$ and H. Fix α with $0 < \alpha < 1$. Since ρ is a positive compact operator of trace 1, there are orthonormal eigenvectors $v_1, v_2, \ldots$ (finitely or infinitely many) whose eigenvalues $\eta_1 \geq \eta_2 \geq \cdots > 0$ are the nonzero eigenvalues of ρ, repeated according to multiplicity, with $\sum_i \eta_i = 1$. Write $\eta_{\leq k} = \eta_1 + \cdots + \eta_k$, with $\eta_{\leq 0} = 0$, and choose k with $\eta_{\leq k} < \alpha \leq \eta_{\leq k+1}$. On the unit vectors of $W = \{v_1, \ldots, v_{k+1}\}^\perp$, the quantity $\langle \rho x \mid x \rangle$ has infimum 0: either W meets the kernel of ρ in a nonzero vector, or the eigenvalues η_i with $i > k+1$ tend to 0. Choose a unit vector $u \in W$ with $\langle \rho u \mid u \rangle < \alpha - \eta_{\leq k}$. For $0 \leq \vartheta \leq 1$, let w_ϑ be the unit vector $\sqrt{1-\vartheta}\, v_{k+1} + \sqrt{\vartheta}\, u$, and let S_ϑ be the subspace spanned by $v_1, \ldots, v_k$ and w_ϑ. Since $\rho v_{k+1} = \eta_{k+1} v_{k+1}$ and u is orthogonal to $v_1, \ldots, v_{k+1}$, the cross terms vanish, and

$$\operatorname{Tr}(\rho P_{S_\vartheta}) = \eta_{\leq k} + (1-\vartheta)\eta_{k+1} + \vartheta \langle \rho u \mid u \rangle,$$

a continuous function of ϑ that equals $\eta_{\leq k+1} \geq \alpha$ at $\vartheta = 0$ and is smaller than α at $\vartheta = 1$. By the intermediate value theorem, it takes the value α. ☺

Proof of Theorem B. In a standard model of T, Theorem A gives $\mu(a) = \operatorname{Tr}(\rho P_a)$ for some density operator ρ on H, and $a \mapsto S_a$ is onto $L(H)$; by Lemma 8.4, the range of μ is $[0,1]$. Since B meets $[0,1]$, some element a satisfies $\mu(a) \in B$, and the value of $\mathbf{1}_B(\mu(a))$ at that element is 1; hence the axiom of Theorem B fails. ☺

9. The proof of Theorem C

The proof of Theorem C, which says that no model of T carries a measure with only the values 0 and 1, works in an arbitrary model of T, standard or not, and uses, besides the basis (e_n), only the eighteen vectors of Table 1. Throughout this section, $\mathcal{M}$ is a model of T in which μ takes only the values 0 and 1; we derive a contradiction, which proves Theorem C. Let $\bar{E}$ and $\bar{P}_a$ be as in Lemma 6.3. By (H2), the vectors e_n form an orthonormal basis of $\bar{E}$: Parseval's identity puts every vector of B, and hence of $\bar{E}$, in their closed span. The proof rests on three facts that hold in every model of T, standard or not.

Lemma 9.1. *(1) For every nonzero x in the Hilbert sort and every $\lambda \in \mathbb{Q}(i)$ with $0 < |\lambda| \leq 1$, we have $\operatorname{at}(\lambda x) = \operatorname{at}(x)$, and $\bar{P}_{\operatorname{at}(x)}$ is the projection onto $\mathbb{C}x$.*

(2) Let s be an element of the sequence sort, and let F be the closed span of its entries in $\bar{E}$. If every e_n lies in F or in $F^\perp$, then $\bar{P}_{\operatorname{span}(s)}$ is the projection onto F.

(3) The measure satisfies $\mu(0) = 0$, and $\operatorname{span}(\varepsilon) = 1$.

Proof. (1) By (At1) and continuity, the range of $\bar{P}_{\operatorname{at}(x)}$ contains x, and any two of its vectors are linearly dependent; hence it is $\mathbb{C}x$, and likewise for λx. The two projections coincide, and axiom (P4) gives $\operatorname{at}(\lambda x) = \operatorname{at}(x)$.

(2) By (Sp1) and linearity, the projection $\bar{P}_{\operatorname{span}(s)}$ fixes every entry of s; hence its range contains F. If $e_n \in F^\perp$, then the vector e_n is orthogonal to every entry, and axiom (Sp2) gives $\bar{P}_{\operatorname{span}(s)} e_n = 0$. The vectors e_n that lie in $F^\perp$ span a dense subspace of $F^\perp$, because the

e_n form an orthonormal basis and each lies in F or in $F^\perp$; by continuity, $\bar{P}_{\mathrm{span}(s)}$ vanishes on $F^\perp$, and its range is F.

(3) For $s = 0$, every vector of the Hilbert sort is orthogonal to the entries, and axiom (Sp2) gives $\bar{P}_{\mathrm{span}(0)} = 0 = \bar{P}_0$; axiom (P4) gives $\mathrm{span}(0) = 0$, and the argument of Step 1 in the proof of Theorem A gives $\mu(0) = 0$. For $s = \varepsilon$, the entries are $2^{-n-1}e_n$ by (K4); the space F is then $\bar{E}$, and part (2) gives $\bar{P}_{\mathrm{span}(\varepsilon)} = \mathrm{Id} = \bar{P}_1$, whence $\mathrm{span}(\varepsilon) = 1$ by (P4). ☺

basis	vectors (coordinates with respect to g_0, g_1, g_2, g_3)			
1	$(0,0,0,1)$	$(0,0,1,0)$	$(1,1,0,0)$	$(1,-1,0,0)$
2	$(0,0,0,1)$	$(0,1,0,0)$	$(1,0,1,0)$	$(1,0,-1,0)$
3	$(1,-1,1,-1)$	$(1,-1,-1,1)$	$(1,1,0,0)$	$(0,0,1,1)$
4	$(1,-1,1,-1)$	$(1,1,1,1)$	$(1,0,-1,0)$	$(0,1,0,-1)$
5	$(0,0,1,0)$	$(0,1,0,0)$	$(1,0,0,1)$	$(1,0,0,-1)$
6	$(1,-1,-1,1)$	$(1,1,1,1)$	$(1,0,0,-1)$	$(0,1,-1,0)$
7	$(1,1,-1,1)$	$(1,1,1,-1)$	$(1,-1,0,0)$	$(0,0,1,1)$
8	$(1,1,-1,1)$	$(-1,1,1,1)$	$(1,0,1,0)$	$(0,1,0,-1)$
9	$(1,1,1,-1)$	$(-1,1,1,1)$	$(1,0,0,1)$	$(0,1,-1,0)$

TABLE 1. The configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]: nine orthogonal bases of a four-dimensional space, formed by eighteen vectors, each of which occurs in exactly two bases.

Proof of Theorem C. The entries of ε are mutually orthogonal, whence $\theta(\varepsilon) = 0$, and Lemma 9.1(1) gives $\mathrm{at}(2^{-n-1}e_n) = \mathrm{at}(e_n)$; by (M2), (K4), (M3) and Lemma 9.1(3), we obtain

$$1 = \mu(1) = \mu(\mathrm{span}(\varepsilon)) = \sum_n \mu(\mathrm{at}(e_n)).$$

As μ takes only the values 0 and 1, there is exactly one $n_0 \in \mathbb{N}$ with $\mu(\mathrm{at}(e_{n_0})) = 1$. Let $j \in \mathbb{N}$ satisfy $4j \le n_0 < 4j + 4$, put $g_i = e_{4j+i}$ for $i = 0, 1, 2, 3$, and let F be the span of $g_0, \dots, g_3$.

Let $\mathcal{B}$ be an orthogonal basis of F consisting of vectors $v_0, \dots, v_3$ whose coordinates with respect to $g_0, \dots, g_3$ lie in $\{0, 1, -1\}$. Each vector $\frac{1}{4}v_i$ has coordinates whose absolute values sum to at most 1, and it is therefore denoted by a term built from the constants e_n by the functions $\lambda x + \lambda' y$. Hence the sequence $\sum_{i<4} \iota_i(\frac{1}{16}v_i)$ is an element $s_{\mathcal{B}}$ of the sequence sort, denoted by the term $\frac{1}{2}\big(\frac{1}{2}\iota_0(\frac{1}{4}v_0) + \frac{1}{2}\iota_1(\frac{1}{4}v_1)\big) + \frac{1}{2}\big(\frac{1}{2}\iota_2(\frac{1}{4}v_2) + \frac{1}{2}\iota_3(\frac{1}{4}v_3)\big)$. The entries of $s_{\mathcal{B}}$ are $\frac{1}{16}v_0, \dots, \frac{1}{16}v_3$ and 0, which are mutually orthogonal; hence $\theta(s_{\mathcal{B}}) = 0$. Each e_n lies in F or in $F^\perp$, and Lemma 9.1(2) shows that $\bar{P}_{\mathrm{span}(s_{\mathcal{B}})}$ is the projection onto F. For a vector v with coordinates in $\{0, 1, -1\}$, we write $\mathrm{at}(v)$ for $\mathrm{at}(\frac{1}{16}v)$; by Lemma 9.1(1), this element does not change when the sign of v is changed. By (P4), all the elements $\mathrm{span}(s_{\mathcal{B}})$ are equal to one element a_F of the Solèr sort, and axiom (M3), Lemma 9.1(1) and $\mu(0) = 0$ give

$$\mu(a_F) = \sum_{v \in \mathcal{B}} \mu(\mathrm{at}(v)).$$

For the basis $g_0, \dots, g_3$ this sum contains $\mu(\mathrm{at}(e_{n_0})) = 1$ and three zeros; hence $\mu(a_F) = 1$, and every such basis $\mathcal{B}$ contains exactly one vector v with $\mu(\mathrm{at}(v)) = 1$.

Apply this to the nine bases of Table 1, whose vectors span pairwise distinct rays. Count the pairs $(\mathcal{B}, v)$ in which $\mathcal{B}$ is one of the nine bases, $v \in \mathcal{B}$ and $\mu(\mathrm{at}(v)) = 1$. Each basis contributes exactly one pair, and the count is 9. On the other hand, each of the eighteen vectors occurs in exactly two bases, and each vector v with $\mu(\mathrm{at}(v)) = 1$ therefore contributes exactly two pairs; the count is even. This contradiction proves the theorem. ☻

Combined with axiom (M1), Theorem C gives the following.

Corollary 9.2 (No hidden variables). *In every model of T, standard or not, some element a of the Solèr sort satisfies $0 < \mu(a) < 1$.*

Proof. By (M1), the measure takes values in $[0, 1]$, and by Theorem C it does not take only the values 0 and 1. ☻

10. The model existence game and finite obstructions

By Theorem C, no model of the theory T of Theorem A, standard or not, carries a measure that takes only the values 0 and 1. In first-order logic, the compactness theorem would turn this into a finite obstruction: some finite part of the theory would already have no models, as in the Kochen–Specker theorem [KS67], in which finitely many vectors suffice. Real-valued $L_{\omega_1,\omega}$ is not compact, but it satisfies a model existence theorem, which extends the classical one for $L_{\omega_1,\omega}$ (see Keisler [Kei71]) and was proved for the logic of Section 5 by the first author [Aub]; for $[0, 1]$-valued $L_{\omega_1,\omega}$, Eagle [Eag14] proved the omitting types theorem by a Baire category argument. In this section we state the model existence theorem as a game between two players: Olivia, who claims that a given theory has no models, and Josh, who claims that it has one. The game follows Väänänen's model existence game for two-valued logic [Vää11]. We show that a theory has no models exactly when Olivia has a winning strategy, and that every winning strategy of Olivia determines a well-founded tree whose leaves are finite obstructions. This tree branches infinitely only at challenges that concern countable suprema and infima, and once Josh's answers to those challenges are fixed, every play ends within a bounded number of rounds. For the theory of Theorem C, the finite obstructions are configurations of the Kochen–Specker type. The section can be read after Section 5; Figure 5 shows a few rounds of the game.

10.1. **The setting.** Until the last subsection, $\mathcal{L}$ is a countable many-sorted vocabulary for the logic of Section 5, and T is a nonempty countable theory in $\mathcal{L}$; in the last subsection they are again the vocabulary of Section 5.2 and the theory of Theorem A. As usual, every sort of a structure is nonempty.

Each predicate symbol P comes with a positive integer r_P such that P takes its values in $[-r_P, r_P]$. By induction on formulas, each formula φ receives a positive rational number r_φ such that φ takes its values in the interval $J_\varphi = [-r_\varphi, r_\varphi]$ in every structure: $r_{P(\bar{t})} = r_P$; for a continuous connective f, the number $r_{f(\varphi_1,\dots,\varphi_\ell)}$ is the least positive integer that is at least the maximum of $|f|$ on the box $J_{\varphi_1} \times \cdots \times J_{\varphi_\ell}$; $r_{\sup_x \varphi} = r_{\inf_x \varphi} = r_\varphi$; and $r_{\sup_n \varphi_n} = r_{\inf_n \varphi_n}$ is the rational uniform bound that comes with the family $(\varphi_n)_{n\in\mathbb{N}}$. These bounds do not change when terms are substituted for free variables. Replacing a condition $\varphi = 0$ by $|\varphi| \le 0$, and then a condition $\varphi \le 0$ by $\sup_{\bar{x}} \varphi \le 0$, where $\bar{x}$ lists the free variables of φ, we may assume that every axiom of T is a condition $\psi \le 0$ for a sentence ψ.

To name the elements of a structure under construction, we add to $\mathcal{L}$, for each sort $\mathbf{S}$, a set $\mathcal{W}_\mathbf{S} = \{w_{\mathbf{S},i} : i \in \mathbb{N}\}$ of new constants of sort $\mathbf{S}$, the *witnesses*; we write $\mathcal{W}$ for the union of the sets $\mathcal{W}_\mathbf{S}$, and $\mathcal{L}(\mathcal{W})$ for the vocabulary obtained by adding the witnesses to $\mathcal{L}$. We use the letters w and w' for witnesses. Constant symbols of $\mathcal{L}$ are treated as function symbols

with no arguments, so that the rules below handle them together with the other function symbols. A sentence of $\mathcal{L}(\mathcal{W})$ is *basic* when it has the form $P(w_1, \dots, w_q)$, where P is a predicate symbol and $w_1, \dots, w_q$ are witnesses.

The players build a partial description of a structure whose elements are named by witnesses. Such a description consists of commitments, which say that the value of a sentence lies in a given interval, and records, which say what a function symbol does to the elements named by given witnesses.

Definition 10.1. A *commitment* is an expression $\varphi \in I$, where φ is a sentence of $\mathcal{L}(\mathcal{W})$ and $I = [\alpha, \beta]$ is an interval with rational endpoints $\alpha \leq \beta$; the interval may be a single point. A *record* is an expression $F(\bar{w}) \mapsto w'$, where F is a function symbol and $\bar{w}, w'$ are witnesses of the appropriate sorts. A *position* is a finite set $\mathfrak{p}$ of commitments and records that contains at most one record $F(\bar{w}) \mapsto w'$ for each function symbol F and each tuple $\bar{w}$. A witness *occurs* in $\mathfrak{p}$ when it occurs in a commitment or a record of $\mathfrak{p}$. A position $\mathfrak{p}$ is *infeasible* when there is a basic sentence $P(\bar{w})$ such that the intervals I for which $\mathfrak{p}$ contains the commitment $P(\bar{w}) \in I$, together with $[-r_P, r_P]$, have empty intersection.

An infeasible position is contradictory in the most elementary way: it assigns to a basic sentence $P(\bar{w})$ finitely many rational intervals that have no common point in $[-r_P, r_P]$ (see Figure 7).

The rules for continuous connectives need two pieces of notation: dyadic grids, and a modulus of uniform continuity for each connective. For $m \in \mathbb{N}$, the *dyadic grid* $\mathcal{G}_m$ is the set of intervals $[i2^{-m}, (i+1)2^{-m}]$, for $i \in \mathbb{Z}$. Let $f\colon \mathbb{R}^\ell \to \mathbb{R}$ be a continuous connective, let $\varphi_1, \dots, \varphi_\ell$ be sentences, and let $k \in \mathbb{N}$. The function f is uniformly continuous on the compact box $J_{\varphi_1} \times \cdots \times J_{\varphi_\ell}$; we write $m(f, \bar{\varphi}, k)$ for the least $m \in \mathbb{N}$ such that $|f(\xi) - f(\xi')| \leq 2^{-k}$ for all points ξ, ξ' of the box with $|\xi_j - \xi'_j| \leq 2^{-m}$ for $j = 1, \dots, \ell$. For an interval $I = [\alpha, \beta]$ and $k \in \mathbb{N}$, we write $I^{(k)} = [\alpha - 2^{-k}, \beta + 2^{-k}]$.

Definition 10.2 (The model existence game). The game $\mathrm{G}(T)$ is played by Olivia and Josh. It starts at the empty position; at each round, Olivia makes a *challenge* and Josh *answers* it by enlarging the current position $\mathfrak{p}$. When Josh answers a challenge, the *first fresh witness* of a sort $\mathbf{S}$ is the witness $w_{\mathbf{S},i}$ with the least index i that occurs neither in $\mathfrak{p}$ nor in the challenge. In each rule other than (Ax) and (Tot), the commitment named by Olivia belongs to $\mathfrak{p}$, and its interval is written $I = [\alpha, \beta]$.

(Ax): Olivia names an axiom $\psi \leq 0$ of T. Josh adds $\psi \in [-r_\psi, 0]$.

(Tot): Olivia names a function symbol F and witnesses $\bar{w}$ of the appropriate sorts. If $\mathfrak{p}$ contains no record $F(\bar{w}) \mapsto w'$, Josh adds such a record, choosing for w' a witness of the sort of the values of F that occurs in $\mathfrak{p}$ or is the first fresh witness of that sort; otherwise he adds nothing.

(Tm): Olivia names a commitment $P(\bar{t}) \in I$, where P is a predicate symbol and $\bar{t}$ is a tuple of terms, together with a term $F(\bar{w})$ that occurs in $\bar{t}$ and in which $\bar{w}$ is a tuple of witnesses (possibly empty). Josh first answers as in (Tot) for F and $\bar{w}$; then, with w' the witness such that the enlarged position contains the record $F(\bar{w}) \mapsto w'$, he adds $P(\bar{t}') \in I$, where $\bar{t}'$ is obtained from $\bar{t}$ by replacing every occurrence of $F(\bar{w})$ by w'.

(Cn): Olivia names a commitment $f(\varphi_1, \dots, \varphi_\ell) \in I$, where f is a continuous connective, and a number $k \in \mathbb{N}$. With $m = m(f, \bar{\varphi}, k)$, Josh chooses intervals $D_1, \dots, D_\ell$ of $\mathcal{G}_m$ such that each D_j meets J_{φ_j} and f maps $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ into $I^{(k)}$, and adds the commitments $\varphi_j \in D_j \cap J_{\varphi_j}$ (see Figure 6).

($\sup \leq$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [-r_\varphi, \beta]$.

(sup ≥): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness $w \in \mathcal{W}_{\mathbf{S}}$ that occurs in $\mathfrak{p}$ or is the first fresh witness of sort $\mathbf{S}$, and adds $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$.

(inf ≥): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [\alpha, r_\varphi]$.

(inf ≤): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness w as in (sup ≥), and adds $\varphi(w) \in [-r_\varphi, \beta + 2^{-k}]$.

($\sup_n \leq$): Olivia names a commitment $\sup_n \varphi_n \in I$, where $(\varphi_n)_{n\in\mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\sup_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [-r, \beta]$.

($\sup_n \geq$): Olivia names a commitment $\sup_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [\alpha - 2^{-k}, r]$.

($\inf_n \geq$): Olivia names a commitment $\inf_n \varphi_n \in I$, where $(\varphi_n)_{n\in\mathbb{N}}$ is a family of formulas with common rational bound $r = r_{\inf_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [\alpha, r]$.

($\inf_n \leq$): Olivia names a commitment $\inf_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [-r, \beta + 2^{-k}]$.

An answer is *legal* when it obeys the rule and every interval that it adds is nonempty, that is, has its left endpoint at most its right endpoint. Josh *loses* at a round if he has no legal answer to Olivia's challenge, or if his answer makes the position infeasible. Olivia wins a play if Josh loses at some round; Josh wins a play if he never loses, in which case the play is infinite.

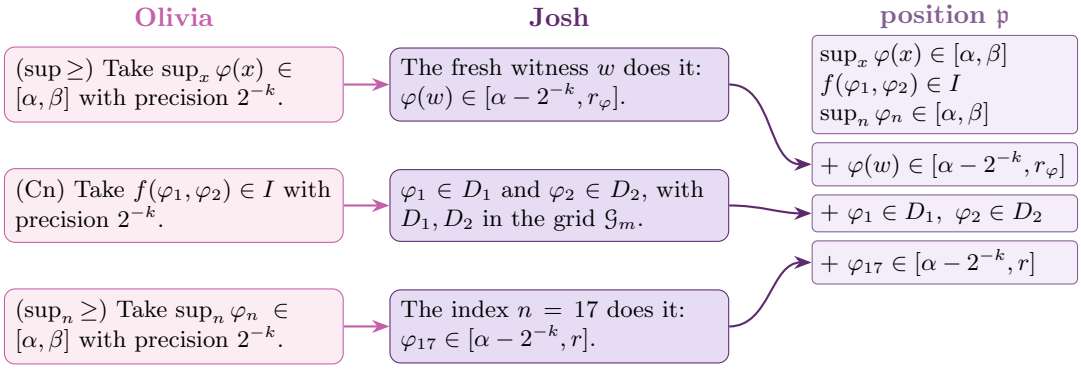


FIGURE 5. Three rounds of the game. Olivia's challenges are on the left, Josh's answers on the right; the position, on the far right, grows by one card at each answer. The first round concerns a quantifier, the second a connective and the third a countable supremum.

Josh's answers are of three kinds, according to the number of choices that the rules leave him. Under the rules (Ax), (sup ≤), (inf ≥), ($\sup_n \leq$) and ($\inf_n \geq$), Josh has no choice: the answer is determined by the challenge. Under the rules (Tot), (Tm), (Cn), (sup ≥) and (inf ≤), Josh has finitely many choices, since only finitely many witnesses occur in a position and only finitely many intervals of $\mathcal{G}_m$ meet a bounded interval. Only under the rules ($\sup_n \geq$) and ($\inf_n \leq$), which concern countable suprema and infima, does Josh choose among infinitely many answers. We call the challenges made under these two rules *countable challenges*, and Josh's answers to them *countable answers*.

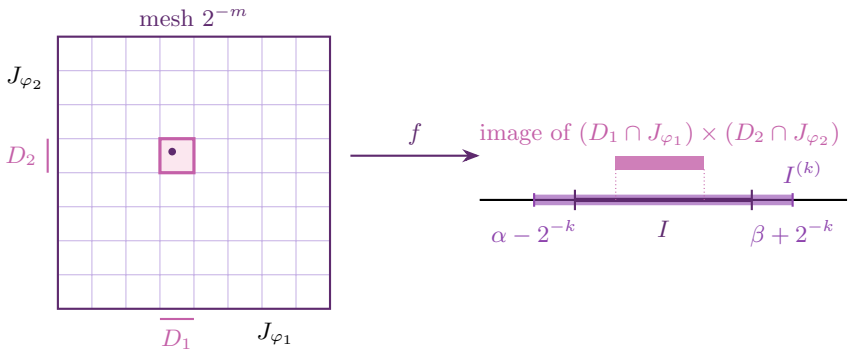


FIGURE 6. The rule (Cn) for a connective f of two variables. Josh chooses a cell $D_1 \times D_2$ of the dyadic grid of mesh 2^{-m} that contains the true values; by uniform continuity, f maps the whole cell into the widened interval $I^{(k)}$.

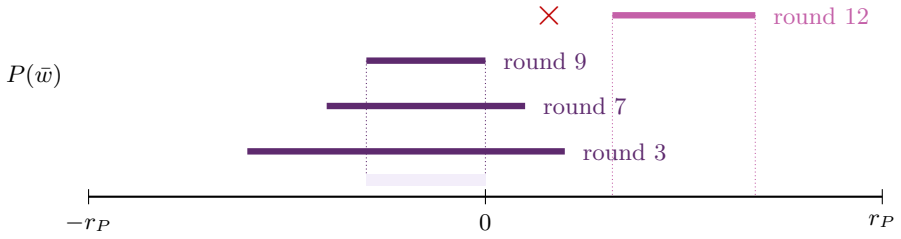


FIGURE 7. An infeasible position: at different rounds, Josh has committed the basic sentence $P(\bar{w})$ to intervals that have no common point.

10.2. **The model existence theorem as a game.** A challenge is *available* at a round when the rules allow Olivia to make it there, that is, when the commitment that it names, if any, belongs to the position at the start of that round. Every sentence that occurs in a play is an atomic sentence of $\mathcal{L}(\mathcal{W})$ or is obtained from a subformula of an axiom of T by substituting closed terms of $\mathcal{L}(\mathcal{W})$ for its free variables; hence only countably many challenges are available at each round, and there is at least one, since T is nonempty. A play is *fair* if every challenge that is available at some round is made by Olivia at that round or at a later one. Olivia has a *fair strategy*, one under which every play is fair, whatever Josh does: she fixes a bijection $i \mapsto (j(i), l(i))$ from $\mathbb{N}$ onto $\mathbb{N} \times \mathbb{N}$ with $j(i) \leq i$ for every $i \in \mathbb{N}$; at each round she enumerates, with repetitions if necessary, the challenges available at that round; and at round i she makes the $l(i)$-th challenge of her enumeration for round $j(i)$. Challenges available at a round remain available later, since positions only grow.

Theorem 10.3 (The model existence theorem as a game). *Let T be a nonempty countable theory of real-valued $L_{\omega_1,\omega}$ in a countable vocabulary, whose axioms are conditions $\psi \leq 0$ for sentences ψ, and let $\mathrm{G}(T)$ be the game of Definition 10.2. The following conditions are equivalent:*

(1) the theory T has a model;
(2) Josh has a winning strategy in $\mathrm{G}(T)$;
(3) Olivia has no winning strategy in $\mathrm{G}(T)$.

Proof. (1) implies (2). Let $\mathcal{N}$ be a model of T. Josh keeps, along the play, a map h from the witnesses occurring in the current position to $\mathcal{N}$ such that every commitment of the position holds in $\mathcal{N}$ when each witness w is interpreted as $h(w)$, and $F^{\mathcal{N}}(h(\bar{w})) = h(w')$ for every record $F(\bar{w}) \mapsto w'$ of the position. When a challenge names a witness that does not yet occur, Josh first extends h to it arbitrarily; this is possible because the sorts of $\mathcal{N}$ are nonempty. He answers as follows. For (Ax), the added commitment holds because $\mathcal{N}$ is a model of T. For (Tot) and (Tm), if the position contains no record $F(\bar{w}) \mapsto w'$, Josh adds one with w' the first fresh witness, and puts $h(w') = F^{\mathcal{N}}(h(\bar{w}))$; in either case, the value of $P(\bar{t}')$ in $\mathcal{N}$ equals that of $P(\bar{t})$. For (Cn), let ξ_j be the value of φ_j in $\mathcal{N}$; Josh chooses $D_j \in \mathcal{G}_m$ containing ξ_j. Every point of $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ has coordinates within 2^{-m} of those of $(\xi_1, \dots, \xi_\ell)$, so f maps it to within 2^{-k} of $f(\xi_1, \dots, \xi_\ell)$, which lies in I. For $(\sup \leq)$, the value of $\varphi(w)$ is at most the supremum, hence at most β. For $(\sup \geq)$, the supremum is at least α; Josh takes the first fresh witness w and lets $h(w)$ be an element of $\mathcal{N}$ at which φ takes a value at least $\alpha - 2^{-k}$. The remaining rules are handled in the same way; for instance, for $(\sup_n \geq)$, some n gives φ_n a value at least $\alpha - 2^{-k}$. Every answer is legal, since the true values lie in the added intervals, and no position is infeasible, since the value in $\mathcal{N}$ of a basic sentence $P(\bar{w})$ lies in every interval committed for it and in $[-r_P, r_P]$. Thus Josh never loses.

(2) implies (1). Let Josh play a winning strategy against a fair strategy of Olivia. The resulting play is infinite and fair; let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots$ be its positions, indexed by $i \in \mathbb{N}$, and $\mathfrak{p}_\infty$ their union. We define a structure $\mathcal{M}$ for $\mathcal{L}(\mathcal{W})$ whose elements are the witnesses: the universe of sort $\mathbf{S}$ is $\mathcal{W}_{\mathbf{S}}$, and each witness is interpreted as itself. For a function symbol F and witnesses $\bar{w}$, Olivia makes the challenge (Tot) for F and $\bar{w}$ at some round, because the play is fair; hence $\mathfrak{p}_\infty$ contains a record $F(\bar{w}) \mapsto w'$, which is unique because every $\mathfrak{p}_i$ is a position, and we put $F^{\mathcal{M}}(\bar{w}) = w'$. For a basic sentence $P(\bar{w})$, the intervals I for which $\mathfrak{p}_\infty$ contains $P(\bar{w}) \in I$, together with $[-r_P, r_P]$, are compact, and every finite subfamily has a common point because no position $\mathfrak{p}_i$ is infeasible; by the finite intersection property, the whole family has a common point, and we let $P^{\mathcal{M}}(\bar{w})$ be one. No continuity is required of $F^{\mathcal{M}}$ or $P^{\mathcal{M}}$ (see Section 5); hence $\mathcal{M}$ is a structure for $\mathcal{L}(\mathcal{W})$.

We claim that every commitment $\varphi \in I$ of $\mathfrak{p}_\infty$ holds in $\mathcal{M}$, and prove it by induction on the construction of φ and, for atomic sentences, on the number of occurrences of function symbols. For a basic sentence, the claim holds by the choice of $P^{\mathcal{M}}$. A non-basic atomic sentence $P(\bar{t})$ contains a term $F(\bar{w})$ whose arguments $\bar{w}$ are witnesses (possibly none, when F is a constant symbol); by fairness, Olivia made the challenge (Tm) for the commitment and this term, and Josh added $P(\bar{t}') \in I$, which holds by the induction hypothesis; since $F^{\mathcal{M}}(\bar{w})$ is the witness w' that replaces $F(\bar{w})$, the sentences $P(\bar{t})$ and $P(\bar{t}')$ have the same value in $\mathcal{M}$. For $f(\varphi_1, \dots, \varphi_\ell) \in I$, fairness gives, for every $k \in \mathbb{N}$, intervals $D_1, \dots, D_\ell$ such that $\mathfrak{p}_\infty$ contains the commitments $\varphi_j \in D_j \cap J_{\varphi_j}$ and f maps $(D_1 \cap J_{\varphi_1}) \times \cdots \times (D_\ell \cap J_{\varphi_\ell})$ into $I^{(k)}$; by the induction hypothesis, the value of $f(\varphi_1, \dots, \varphi_\ell)$ in $\mathcal{M}$ lies in $I^{(k)}$ for every k, hence in I. For $\sup_x \varphi(x) \in [\alpha, \beta]$, fairness puts in $\mathfrak{p}_\infty$ a commitment $\varphi(w) \in [-r_\varphi, \beta]$ for every witness w of the sort of x, that is, for every element of $\mathcal{M}$ of that sort, and, for every $k \in \mathbb{N}$, a commitment $\varphi(w) \in [\alpha - 2^{-k}, r_\varphi]$ for some witness w; by the induction hypothesis, the supremum lies in $[\alpha, \beta]$. The remaining cases are similar. Every axiom $\psi \leq 0$ of T was named in a challenge (Ax), so $\mathfrak{p}_\infty$ contains $\psi \in [-r_\psi, 0]$, and $\mathcal{M}$ satisfies $\psi \leq 0$.

It remains to pass to a reduced structure. For elements a, a' of $\mathcal{M}$ of the same sort, let $a \sim a'$ when $\chi(a, \bar{e})$ and $\chi(a', \bar{e})$ have the same value for every atomic formula $\chi(x, \bar{y})$ and all parameters $\bar{e}$. If $a \sim a'$, then $F(a, \bar{e}') \sim F(a', \bar{e}')$ for every function symbol F and all parameters $\bar{e}'$, since $\chi(F(x, \bar{y}'), \bar{y})$ is again atomic; by transitivity, $\sim$ is a congruence for every

function symbol, whether or not it is continuous. The quotient $\mathcal{M}/\!\sim$ is reduced, and the quotient map is surjective and preserves the value of every atomic formula; by induction on formulas, it preserves the value of every formula. Hence the reduct of $\mathcal{M}/\!\sim$ to $\mathcal{L}$ is a model of T.

(2) and (3) are equivalent. Regard $\mathrm{G}(T)$ as an infinite game in which, after Josh has lost, the play continues with arbitrary moves. Olivia wins exactly when Josh loses at some finite round; in the language of descriptive set theory, the set of her wins is *open* in the space of infinite sequences of moves, whose basic open sets consist of the sequences that extend a given finite one. The Gale–Stewart theorem [GS53] says that an infinite game of perfect information in which the set of wins of one player is open is *determined*: one of the two players has a winning strategy. Since the two players cannot both have winning strategies, (2) and (3) are equivalent.

Remark 10.4 (Baire category and the Banach–Mazur game). Let σ' be a winning strategy for Josh, and let $X_{\sigma'}$ be the set of infinite plays in which Josh follows σ', with the topology whose basic open sets consist of the plays that extend a given finite play. Since the moves range over countable sets, $X_{\sigma'}$ is homeomorphic to a closed subspace of the Baire space $\mathbb{N}^{\mathbb{N}}$, the space of sequences of natural numbers with the product of the discrete topologies. It is therefore a *Polish* space, that is, separable and completely metrizable, and in particular *Čech-complete*, which is the generality in which the Baire category theorem holds. It is not compact, because at most rounds Olivia has infinitely many challenges to choose from (a witness, or a number k or n). In the *Banach–Mazur game* on a space, two players alternately choose nonempty open sets, each contained in the one before, and the second player wins if the intersection of the chosen sets lies in a given set. For each challenge and each round, the set of plays in which the challenge, if available at that round, is made at that round or later is open and dense in $X_{\sigma'}$; hence the fair plays form a dense G_δ subset of $X_{\sigma'}$, a countable intersection of dense open sets, which is nonempty by the Baire category theorem. Equivalently, in the Banach–Mazur game on $X_{\sigma'}$ in which the second player wins if the resulting play is fair, Olivia's bookkeeping is a winning strategy for the second player (compare Oxtoby [Oxt57]). This is the analogue of the Baire category argument by which Eagle [Eag14] proves the omitting types theorem for $[0,1]$-valued $L_{\omega_1,\omega}$; in the proof of Theorem 10.3 it reduces to Olivia's bookkeeping, because only countably many challenges are available at each round (see Figure 8).

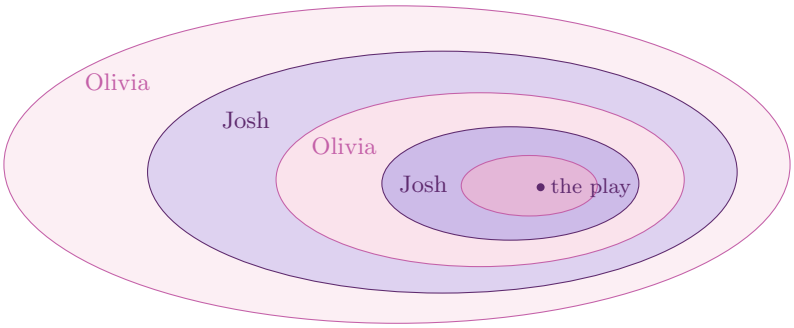


FIGURE 8. The Banach–Mazur view of a play: each move shrinks the set of plays that extend the current finite play, alternately by Olivia (pink) and by Josh (purple), and the nested sets converge to a single play.

10.3. **Finite obstructions.** A winning strategy for Olivia is a certificate that a theory has no models, and its shape shows where finite obstructions suffice. Let σ be a strategy for Olivia. The *tree of* σ, denoted $\mathcal{P}_\sigma$, is the set of finite plays in which Olivia follows σ and Josh has not lost before the last round, ordered by extension. A finite play of $\mathcal{P}_\sigma$ in which Josh has not lost has as children its extensions by Olivia's next challenge and a legal answer of Josh, and, when Josh has no legal answer to that challenge, its extension by the challenge alone. A *leaf* of $\mathcal{P}_\sigma$ is a finite play with no children, that is, one at which Josh has lost. The strategy σ is winning exactly when $\mathcal{P}_\sigma$ has no infinite branch, that is, when $\mathcal{P}_\sigma$ is *well-founded*. The *rank* of a well-founded tree is defined by recursion: a leaf has rank 0, a node that is not a leaf has as rank the least ordinal greater than the ranks of all its children, and the rank of the tree is the rank of its root. Since $\mathcal{P}_\sigma$ is countably branching, it is countable, and its rank is a countable ordinal; the rank is finite exactly when the plays in which Olivia follows σ have bounded length.

The *countable record* of a node of $\mathcal{P}_\sigma$ is the finite sequence of Josh's countable answers along the branch from the root to the node, in the order in which they were given. The *skeleton* of σ is the set of countable records of nodes of $\mathcal{P}_\sigma$, ordered by extension.

Corollary 10.5 (Finite obstructions). *Let T be a nonempty countable theory of real-valued $L_{\omega_1,\omega}$ that has no models. Then Olivia has a winning strategy in $\mathrm{G}(T)$, and for every winning strategy σ of Olivia, the tree $\mathcal{P}_\sigma$ has the following properties.*

1. *The tree $\mathcal{P}_\sigma$ is well-founded: every play in which Olivia follows σ ends after finitely many rounds, at a leaf.*
2. *At every leaf, the last position $\mathfrak{p}$ is a finite set of commitments and records, involving finitely many axioms of T and finitely many witnesses, that is satisfied in no structure under any interpretation of the witnesses. Either $\mathfrak{p}$ is infeasible, in which case a basic sentence receives finitely many rational intervals with no common point, or Josh has no legal answer to Olivia's last challenge, which names a commitment of $\mathfrak{p}$ that holds in no structure.*
3. *A node of $\mathcal{P}_\sigma$ has infinitely many children only if σ makes a countable challenge at it; every other node has finitely many children.*
4. *If no node of $\mathcal{P}_\sigma$ is the site of a countable challenge, as is the case when no axiom of T contains a countable supremum or infimum, then $\mathcal{P}_\sigma$ is finite. In that case there is a number N_σ such that every play in which Olivia follows σ ends within N_σ rounds, and T has a finite subset that has no models.*
5. *The skeleton of σ is well-founded, and for every element s of the skeleton there is a number $N_\sigma(s)$ such that every play in which Olivia follows σ and Josh's countable answers are, in order, the entries of s, ends within $N_\sigma(s)$ rounds.*

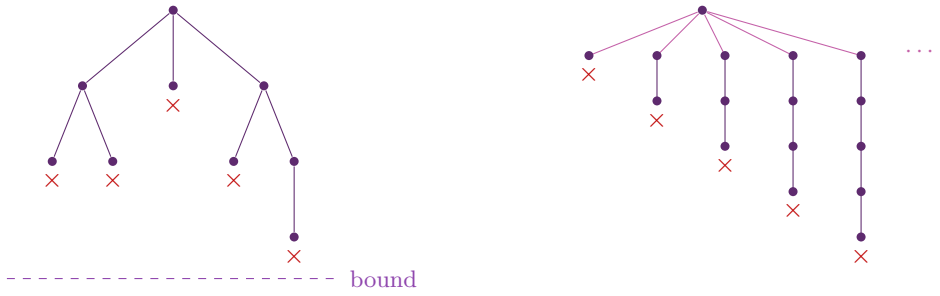
Proof. By Theorem 10.3, Olivia has a winning strategy; let σ be one. Part (1) holds because σ is winning. For part (2), a position contains finitely many commitments and records, and the argument for "(1) implies (2)" in the proof of Theorem 10.3 shows that, at a position satisfied in some structure under some interpretation of the witnesses, the position is not infeasible and Josh has a legal answer to every challenge. Part (3) follows from the count of Josh's possible answers in the paragraph after Definition 10.2.

Parts (4) and (5) rest on König's lemma: an infinite tree in which every node has finitely many children has an infinite branch. For part (4), the tree $\mathcal{P}_\sigma$ is well-founded and, by part (3), finitely branching; hence it is finite, and the length of its longest branch gives N_σ. Let T' be the set of axioms named in challenges (Ax) at nodes of $\mathcal{P}_\sigma$, together with one axiom of T if there are none. In $\mathrm{G}(T')$, every challenge that σ makes at a node of $\mathcal{P}_\sigma$ is allowed by the

rules, and Josh has the same answers as in $\mathrm{G}(T)$; hence σ wins $\mathrm{G}(T')$, and the finite theory T' has no models by Theorem 10.3.

For part (5), let s be a finite sequence of natural numbers, and let $\mathcal{P}_\sigma^s$ be the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of s and whose parent, if it has one, does not already have s as its countable record and a countable challenge as its next move. In $\mathcal{P}_\sigma^s$, a node at which σ makes a countable challenge has at most one child, the one whose countable answer is the next entry of s, unless its countable record is already s, in which case it has no children; by part (3), every node of $\mathcal{P}_\sigma^s$ has finitely many children. Since $\mathcal{P}_\sigma^s$ is well-founded, it is finite by König's lemma, and the length of its longest branch gives $N_\sigma(s)$; every play whose countable answers are the entries of s runs inside $\mathcal{P}_\sigma^s$. If the skeleton had an infinite branch $s_1 \subsetneq s_2 \subsetneq \cdots$, the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of some s_i would form a finitely branching subtree of $\mathcal{P}_\sigma$ containing nodes of every countable record s_i, hence infinite; by König's lemma it would have an infinite branch, contradicting part (1). ☺

Part (4) of Corollary 10.5 gives a proof of the compactness theorem for countable theories of the finitary part of the logic: a countable theory without countable suprema and infima that has no models has a finite subset that has no models. Part (3) shows where compactness can fail: the tree branches infinitely only at countable challenges, where Josh chooses an index n among infinitely many. Part (5) shows that this is the only obstruction to a uniform bound: the skeleton, a well-founded countably branching tree, records the infinitary shape of the obstruction, and below each element of the skeleton the rest of the play is bounded (see Figures 9 and 10). For a theory with countable suprema and infima, the single finite obstruction of the compact case is thus replaced by a well-founded tree of finite obstructions, one at each leaf.



finitely branching: finite

countably branching: no bound

FIGURE 9. Left: a finitely branching tree without infinite branches is finite, by König's lemma, and its plays have bounded length. Right: a countably branching tree without infinite branches may have plays of every finite length, as in this fan with one branch of each length.

10.4. **The Bell–Kochen–Specker obstruction.** We now return to the theory T of Theorem A and its measure μ. Let T^{01} be the theory T together with the axiom $\sup_a \min\big(\mu(a), 1-\mu(a)\big) \leq 0$, the normal form of the axiom of Theorem C, which says that μ takes only the values 0 and 1. By Theorem C, the theory T^{01} has no models, and Corollary 10.5 applies to it.

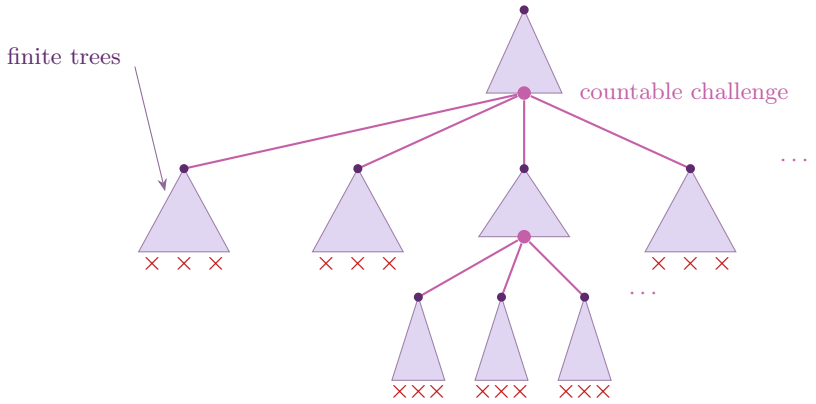


FIGURE 10. The tree of a winning strategy for Olivia. Countable challenges (pink fans) form the skeleton; below each element of the skeleton hangs a finite tree (shaded), whose leaves are finite obstructions (marked ×).

Proposition 10.6. *Let T^{01} be the theory T of Theorem A together with the axiom*

$$\sup_a \min\bigl(\mu(a), 1-\mu(a)\bigr) \le 0.$$

(1) Olivia has a winning strategy in $\mathrm{G}(T^{01})$; every play in which she follows it ends after finitely many rounds, at a position that is satisfied in no structure.
(2) Every finite subset of T^{01} has a model.
(3) The skeleton of every winning strategy of Olivia in $\mathrm{G}(T^{01})$ has more than one element: Olivia cannot win without countable challenges.

Proof. Part (1) is Corollary 10.5(1), since T^{01} has no models by Theorem C. For part (2), let T' be a finite subset of T^{01}, and let $S \subseteq \mathbb{N}$ be the finite set of numbers n for which the instance $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$ of axiom (K4) belongs to T'. In the structure of Remark 11.2 that satisfies every axiom of T except (K4), replace the interpretation of ε by the finitely supported sequence whose entry of index n is $2^{-n-1}e_n$ for $n \in S$ and 0 otherwise. The new sequence lies in the sequence sort of that structure, since its norm is less than 1. The constant ε occurs in no axiom other than (K4), and every other axiom of T holds in that structure whatever element interprets ε; hence the new structure satisfies every axiom of T except the instances of (K4) with $n \notin S$, and its measure takes only the values 0 and 1. It is therefore a model of T'. For part (3), if the skeleton of a winning strategy had only one element, the empty sequence, no node of its tree would be the site of a countable challenge, and by Corollary 10.5(4) the theory T^{01} would have a finite subset without models, contradicting part (2). ☺

Proof of Theorem D. The first three assertions are Theorem 10.3 and Corollary 10.5; the last is Proposition 10.6. ☺

We describe informally a winning strategy for Olivia that follows the proof of Theorem C (see Figures 11 and 12). Recall that the constant ε names the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$ built from the basis $(e_n)_{n\in\mathbb{N}}$ of the Hilbert sort. Olivia first obtains, from the instance of axiom (M3) at ε and from the instances of (H1), (H2), (OL2), (P4), (At1), (K2), (K4), (Sp1), (Sp2), (M1) and (M2) used in the proofs of Lemma 9.1 and Theorem C, commitments saying that the sum of the values $\mu(\mathrm{at}(\kappa_n\varepsilon))$ is close to 1. A countable challenge $(\sup_n \ge)$ to the

supremum $\sup_N \min\bigl(2, \sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))\bigr)$ in (M3) makes Josh name a number N for which the finite sum $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ is close to 1. Challenges to the axiom of T^{01} at witnesses that name the elements $\mathrm{at}(\kappa_n\varepsilon)$, for $n < N$, followed by challenges (Cn) and by challenges to (M1), make Josh commit each value $\mu(\mathrm{at}(\kappa_n\varepsilon))$ to an interval close to 0 or close to 1; since Olivia chooses the precision of these challenges after Josh has named N (finer than $1/(2N)$, say), the finite sum forces at least one $n_0 < N$ with value close to 1. Through (K4) and the identifications behind Lemma 9.1(1), the element $\mathrm{at}(\kappa_{n_0}\varepsilon)$ is $\mathrm{at}(e_{n_0})$. In the block of four basis vectors that contains e_{n_0}, Olivia then challenges the instances of the axioms that the proof of Theorem C uses for the nine bases of Table 1 and for the block itself. Josh's answers commit each of the eighteen values $\mu(\mathrm{at}(v))$ to an interval close to 0 or close to 1, and each of the nine sums over a basis to an interval close to 1; whatever finitely many choices Josh makes, the parity argument in the proof of Theorem C shows that his commitments are incompatible, and Olivia can drive them, through the instances of (OL2), (K2) and (P4) that identify the elements $\mathrm{at}(v)$ reached through different bases, to a basic sentence that receives intervals with no common point. Thus the final position of each leaf contains a finite configuration of the Kochen–Specker type inside one block of four basis vectors. Josh's answer N bounds the block, and his finitely many choices of intervals select it; the countable suprema in (H2), in the formula θ and in the uses of the device (2) in (P4) and (Sp2) contribute further countable challenges along the way.

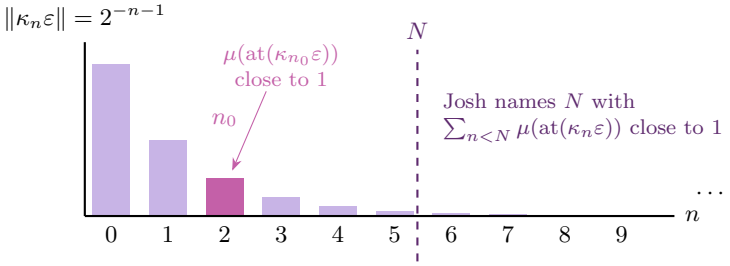


FIGURE 11. The first countable choice. The entries of ε have norms 2^{-n-1}; Josh names N with $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ close to 1, and the two-valued axiom then forces some $n_0 < N$ with value close to 1.

Question 10.7. Does Olivia have a winning strategy in $\mathrm{G}(T^{01})$ whose tree has finite rank, that is, such that every play in which she follows it ends within a fixed number of rounds?

By Proposition 10.6(3), countable challenges cannot be avoided; Question 10.7 asks whether the plays can nevertheless be bounded in length, independently of Josh's countable answers. By Corollary 10.5(5), they are bounded once Josh's countable answers are fixed.

Remark 10.8 (Standard models). The game $\mathrm{G}(T)$ concerns all models of T, standard or not, whereas Theorem B concerns standard models only, and standardness is not expressed by axioms. For a Borel set B that meets $[0,1]$, let T_B be the theory T together with the axiom $\sup_a \mathbf{1}_B(\mu(a)) \le 0$, the normal form of the axiom of Theorem B. Olivia has a winning strategy in $\mathrm{G}(T_B)$ exactly when T_B has no models at all, standard or not. This is the case when B contains 0 or 1, by Lemma 9.1(3) and (M2), and when B contains $(0,1)$, by (M1) and Theorem C; for the remaining sets B the question is open.

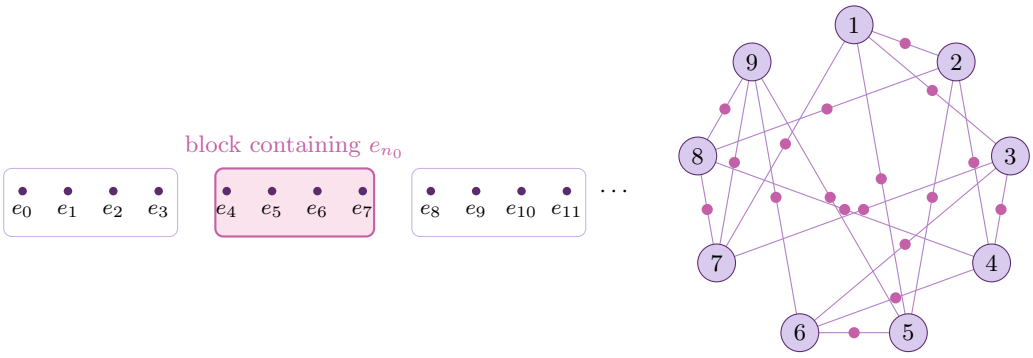


FIGURE 12. The finite obstruction at a leaf. Left: the basis vectors grouped in blocks of four, with the block that contains e_{n_0} highlighted. Right: the nine bases of Table 1 (numbered circles) and its eighteen rays (dots), each ray joined to the two bases that contain it. A two-valued measure would give the value 1 to exactly one ray in each basis; counting these rays basis by basis would count each of them twice, an even number, although there are nine bases.

11. REMARKS

The remarks that follow compare the theory with its predecessor, explain why standardness cannot be dropped from Theorems A and B, and describe the other choices of scalars, the logical form of the axioms and their redundancies.

Remark 11.1 (Solèr triples). The structures considered here refine the Solèr triples of Aubone, Dueñez and Iovino [ADI], in which a Hilbert space and a Solèr lattice are linked by a membership relation. First, in a Solèr triple the lattice is assumed complete, whereas here completeness of the Solèr sort is a consequence of the axioms of T_1 in standard models (see Theorem 8.3). Second, the membership relation is replaced by the projection π: a vector x belongs to a exactly when $\|\frac{1}{2}x - \frac{1}{2}\pi(a,x)\| = 0$. In real-valued logic an existential quantifier becomes an infimum. Read in this way, the condition "$x = y + z$ for some member y of a and some member z of b" holds exactly at the vectors whose distance from $S_a + S_b$ is 0, that is, on the closure of $S_a + S_b$, which is the join of S_a and S_b in $L(H)$. Third, the functions at and rep provide the bijection between the rays of H, which correspond to the unit vectors up to a phase (a complex factor of modulus one), and the atoms of the Solèr sort (see Proposition 7.4).

Remark 11.2 (Nonstandard models). Standardness cannot be dropped from Theorem B or from Theorem A(1): by the downward Löwenheim–Skolem theorem, T has countable models, in which the range of μ is countable and the Solèr sort, being countable, is not of the form $L(H)$. Without the sequence sort, even the conclusion of Gleason's theorem can fail. Let E_0 be the space of finitely supported complex sequences with the inner product of ℓ^2, and let Λ_0 be the lattice of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0 (see Example 4.3). Take the closed unit ball of E_0 as Hilbert sort and Λ_0 as Solèr sort, with the orthogonal projections onto its elements, $\mathrm{at}(x) = \mathbb{C}x$, a choice of rep, and the standard unit vectors as (e_n). The resulting structure is a model of T_0 that is not standard. Every maximal orthonormal family of E_0 is countable and has join 1 in Λ_0; a σ-additive state on Λ_0 would therefore yield a frame function on E_0, a function on the unit vectors whose sum over every maximal orthonormal family is 1 and whose restriction to the unit vectors

of every finite-dimensional subspace is a frame function of that subspace. By a theorem of Dvurečenskij [Dvu95, Theorem 3.1] (compare Hamhalter and Pták [HP87]), an inner product space that carries such a function is complete, and E_0 is not. Hence Λ_0 has no σ-additive states, although density operators on ℓ^2 abound, and the conclusion of Theorem 7.7 fails for this nonstandard model.

The axioms of Theorem B exclude, one Borel set at a time, every range other than $[0,1]$ in standard models; but no sentence says that the range of μ is disconnected. Indeed, suppose that a sentence ψ took the value 0 exactly in the models of T in which the range of μ is disconnected. In a standard model $\mathcal{M}$ of T, the range of μ is $[0,1]$ by Theorem B, and so the value of ψ in $\mathcal{M}$ is not 0. By the downward Löwenheim–Skolem theorem, $\mathcal{M}$ has a countable elementary substructure $\mathcal{N}$ for a countable fragment containing T and ψ; the value of ψ in $\mathcal{N}$ is also not 0, yet the range of μ in $\mathcal{N}$ is countable, and therefore disconnected, a contradiction.

Whether a nonstandard model of T can carry a measure that is not of the form $a \mapsto \operatorname{Tr}(\rho \bar{P}_a)$ we leave open. Axiom (K4) is essential to Theorem C. Take the closed unit ball of ℓ^2 as Hilbert sort, the lattice L_{fc} of Example 7.6, with the same interpretations of d, π, at, rep and (e_n), as Solèr sort, the finitely supported sequences in the closed unit ball of $\ell^2(\mathbb{N}, \ell^2)$ as sequence sort, with ι_n and κ_n as in Theorem 8.3, $\varepsilon = 0$, and $\operatorname{span}(s)$ the span of the entries of s, and let μ vanish on the finite-dimensional subspaces and equal 1 on their orthogonal complements. Every axiom of T except (K4) holds, because the instances of (M3) then concern only finite orthogonal families, and μ takes only the values 0 and 1.

Remark 11.3 (Scalars). The vocabulary fixes the complex numbers as scalars. For real Hilbert spaces, replace $\mathbb{Q}(i)$ by $\mathbb{Q}$ and omit the imaginary parts; for quaternionic spaces, use the quaternions with rational coordinates, acting on the left, and the four real components of the form; in (At4), "complex number of modulus one" becomes a real or quaternionic scalar of modulus one. Each choice yields the analogue of Theorem 8.3 for the corresponding case of Theorem 3.6, and Theorem C holds in all three cases, since the vectors of Table 1 have real coordinates.

Remark 11.4 (The logical form of the axioms). The axioms (H1), (OL), (P1)–(P3), (Sub0), (Sub4), (At1), (At2), (K1), (K2), (K4), (Sp1), (M1) and (M2) are universal conditions built from finitary formulas. Axioms (H2), (Sub2), (Sub3), (K3), (Sp2), (M3) and the formula θ use countable connectives whose arguments are uniformly bounded. Axioms (P4), (Sub1), (At4), (At5), (Sp2), (Sp3) and (M3) use the device (2), which needs only uniform bounds; the functions at, rep and span are not continuous, as Section 5 allows. Axiom (At3) uses the formula $\chi(a)$, which contains a quantifier over the Solèr sort.

Remark 11.5 (Redundancy). In standard models, several axioms follow from others: (Sub1) from (Sub2) and (P4); (Sub3) from (Sub2), (Sub4), (OL) and (P4); (At3) from (At1), (At2) and Proposition 7.3; and (At4) from (At1), (P4) and Proposition 7.3. These axioms are stated nonetheless, so that each property is visible in the axioms.

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